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ADHESIVE COMPOSITION COMPRISING HEAT TREATED DRY PLANT MEAL AND A REACTIVE PREPOLYMER NOT BEING SOLUBLE IN WATER

Abstract

The invention relates to a process for preparing an adhesive composition comprising a heat treated dry plant meal, said heat treated dry plant meal and relating dispersions, adhesive compositions, articles and uses of said heat treated dry plant meal.

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Background/Summary

[0001] The present invention relates to a process for preparing an adhesive composition comprising a heat treated dry plant meal, said heat treated dry plant meal and relating dispersions, adhesive compositions, articles and uses of said heat treated dry plant meal. Adhesive compositions are extensively used in the industry, in particular, the wood product industry, to make composites. These adhesive compositions usually contain harmful chemicals for the environment and for individuals such as, for example, urea formaldehyde or phenol formaldehyde.

[0002] To avoid these harmful chemicals, research is underway to provide more eco-friendly adhesive compositions.

[0003] For example, WO2011156380 describes an adhesive composition comprising a reactive prepolymer and ground plant meal.

[0004] However, the adhesive composition obtained usually have a high viscosity, which may result in processability problems, for example by preventing its use for applications such as spraying.

[0005] Therefore, there is a need for improving processability of eco-friendly adhesive compositions, in particular by decreasing its viscosity, without compromising the final properties of the article made therefrom. Furthermore, there is a need for improving processability of these adhesive compositions without decreasing the plant meal content.

[0006] The inventors have surprisingly found that heat treatment of a dry plant meal before mixing it with a reactive prepolymer improves the processability of the resulting adhesive composition. In particular, at a same plant meal content, the resulting adhesive composition has a lower viscosity than when the plant meal was not heat treated. Furthermore, the inventors have also surprisingly found that a heat treated dry plant meal can be used to stabilize an emulsion, in particular, an adhesive composition comprising a reactive prepolymer not being soluble in water.

[0007] Therefore, in a first aspect, the invention relates to a process for preparing an adhesive composition comprising the steps of: [0008] providing a heat treated dry plant meal which is capable of reducing the viscosity of a dispersion in water of said heat treated dry plant meal compared to the same dispersion in water of the non heat treated dry plant meal, and [0009] mixing the heat treated dry plant meal with water and a reactive prepolymer to provide an adhesive composition, the reactive prepolymer not being soluble in water.

[0010] By “plant meal”, it is meant a meal under a powder (loose), pellet or flake form obtained from a plant or a part thereof such as seeds, which can have been submitted to an extraction step such as, for example, a fat or protein extraction. In particular, the plant meal may be whole plant meal, plant seed meal, defatted plant meal (such as solvent extracted plant meal), press cake (such as hot or cold-press cake), protein isolate, protein concentrate or a mixture thereof. Preferably, the plant meal is a defatted plant meal (such as solvent extracted plant meal), more preferably, the plant meal is a defatted plant seed meal, in particular a defatted oilseed meal.

[0011] Advantageously, the plant meal is not obtained from straw, paper, cardboard nor wood.

[0012] The solvent used to prepare the solvent extracted plant meal may be organic and/or aqueous, such as hexane, alcohol or aqueous alcohol, preferably, hexane or aqueous alcohol, more preferably hexane.

[0013] By “protein isolate”, it is meant a plant meal extract comprising at least 80 wt % proteins based on dry matter.

[0014] By “protein concentrate”, it is meant a plant meal extract comprising at least 60 wt % and less than 80 wt % proteins based on dry matter.

[0015] Advantageously, the plant meal is derived from corn, wheat, sunflower, cotton, castor, soy, camelina, flax, jatropha, mallow, peanuts, sugarcane bagasse, tobacco and/or a plant of the genus

Brassica (such as rape plant, canola, and/or *Brassica carinata*).

[0016] Preferably, the plant meal is derived from corn, flax, soy, sunflower and/or a plant of the genus *Brassica* (such as rape plant, canola and/or *Brassica carinata*).

[0017] More preferably, the plant meal is selected from the group consisting of corn draff, flaxseed meal, soybean meal, sunflower seed meal, rapeseed meal (in particular rapeseed meal pellets), canola seed meal, rapeseed cold press cake, soy protein concentrate, soy protein isolate and *Brassica carinata* meal.

[0018] Still more preferably, the plant meal is selected from the group consisting of corn draff, flaxseed meal, sunflower seed meal, rapeseed seed meal (in particular rapeseed meal pellets), canola seed meal, rapeseed cold press cake and *Brassica carinata* meal.

[0019] Even more preferably, the plant meal is sunflower meal, canola meal or rapeseed meal, in particular, sunflower meal or rapeseed meal, for example, rapeseed meal.

[0020] In a preferred embodiment, the plant meal is sunflower seed meal, canola seed meal or rapeseed seed meal, in particular, rapeseed seed meal.

[0021] Advantageously, the plant meal comprises less than 15 wt % oil, preferably less than 5 wt % oil, more preferably, less than 4 wt % oil, even more preferably, less than 3 wt % oil, based on dry matter.

[0022] By “heat treated dry plant meal”, it is meant that said heat treated dry plant meal is obtained by submitting a dry plant meal to a heat treatment (under dry conditions) at a temperature and during a treatment time sufficient to provide a heat treated dry plant meal which is capable of reducing the viscosity of a dispersion in water of said heat treated dry plant meal compared to the same dispersion in water of the non heat treated dry plant meal.

[0023] Preferably, the dry plant meal is submitted to a heat treatment at atmospheric pressure. In particular, water can go out of the device wherein the dry plant meal is heat treated, allowing decreasing the residual moisture comprised in the dry plant meal.

[0024] Advantageously, the heat treatment is performed at a temperature of at least 100° C., preferably, at least 115° C., more preferably, at least 135° C., even more preferably, at least 150° C.

[0025] Advantageously, the heat treated dry plant meal has a dry content greater than 96 wt %, preferably, greater than 98 wt %, more preferably, comprised between 99 wt % and 100 wt %. The dry content can be determined according to the first method set forth in Example 1.

[0026] By “dry plant meal”, it is meant a plant meal as described above and being under a dry form, i.e. not containing free water, such as not being in solution (e.g., aqueous solution), in particular, a plant meal having a dry content greater than or equal to 80 wt %, preferably, greater than or equal to 85 wt %, more preferably, between 88 wt % and 96 wt %. The dry content can be determined according to the first method set forth in Example 1.

[0027] Therefore, the heat treatment is not performed in a liquid solution (such as an aqueous solution), and no water is added during the heat treatment.

[0028] By “capable of reducing the viscosity”, it is meant that by comparing the viscosity of a first dispersion consisting of a dry plant meal (not subjected to a heat treatment) and water at a particular solid content, and of a second dispersion consisting of the dry plant meal that was heat treated and water at the same solid content and prepared in the same conditions (such as time, temperature, pressure, mixing conditions, for example as described in Example 2 “Preparation of a heat treated dry plant meal dispersion”), the relative viscosity of the second dispersion is less than the relative viscosity of the first dispersion, the latter being set to 100%. Preferably, the relative viscosity of the second dispersion is less than 80% of the first dispersion, more preferably, less than 50%, even more preferably, less than 25%, still more preferably, less than 15%.

[0029] It should be noted that the words “submitted” or “subjected” are used interchangeably throughout all the application.

[0030] By “reactive prepolymer”, it is meant a mixture of monomers and/or oligomers comprising reactive groups.

[0031] By “not being soluble in water”, it is meant that the reactive prepolymer has a water solubility of less than 1 g/L at a temperature of 23° C. In the present application, temperatures are given according to STP.

[0032] Preferably, the reactive prepolymer is an isocyanate-based prepolymer, an epoxy-based prepolymer, a latex prepolymer or a combination thereof.

[0033] More preferably, the reactive prepolymer is an isocyanate-based prepolymer, an epoxy-based prepolymer or a combination thereof.

[0034] Even more preferably, the reactive prepolymer is an isocyanate-based prepolymer.

[0035] By “isocyanate-based prepolymer”, it is meant a reactive prepolymer comprising a reactive isocyanate group.

[0036] By “epoxy-based prepolymer”, it is meant a reactive prepolymer comprising a reactive epoxide group.

[0037] By “latex prepolymer”, it is meant a reactive prepolymer comprising a reactive alkenyl group.

[0038] Advantageously, the isocyanate-based prepolymer is preferably an organic polyisocyanate or a reaction product between an organic polyisocyanate and, for example, a polypeptide, a polyol, an amine-based polyol, an amine containing compound, a hydroxy containing compound or a combination thereof.

[0039] By “organic polyisocyanate”, it is meant an organic compound having several isocyanate groups. In particular, the organic polyisocyanate can have the formula $R(NCO)_n$ wherein n is 2 and R can be an aromatic, a cycloaliphatic, an aliphatic, each having from 2 to about 20 carbon atoms.

[0040] Preferably, the isocyanate-based prepolymer is an aliphatic polyisocyanate comprising an isophorone diisocyanate, 3-isocyanatomethyl-3,5,5-trimethyl-cyclohexyl isocyanate, triphenylmethane-4,4',4''-triisocyanate, polymeric diphenylmethane diisocyanate (pMDI), 4,4'-methylenediphenyl diisocyanate (4,4'-MDI), 2,4'-methylenediphenyl diisocyanate (2,4'-MDI), 2,2'-methylenediphenyl diisocyanate (2,2'-MDI), p-phenylene diisocyanate, m-phenylene diisocyanate, 2,4-toluene diisocyanate, 2,6-toluene diisocyanate, 1,2-naphthalene diisocyanate, 1,5-naphthalene diisocyanate, dianisidine diisocyanate, 3,3'-dimethyl-4,4'-biphenylenediisocyanate, pMDI that has been reacted with a hydroxyl-functional compound such as a polyol and/or saturated analogues of the above-mentioned aromatic isocyanate-based prepolymers.

[0041] More preferably, the isocyanate-based prepolymer is polymeric diphenylmethane diisocyanate (pMDI), 4,4'-methylenediphenyl diisocyanate (4,4'-MDI), 2,4'-methylenediphenyl diisocyanate (2,4'-MDI), 2,2'-methylenediphenyl diisocyanate (2,2'-MDI), p-phenylene diisocyanate, m-phenylene diisocyanate, 2,4-toluene diisocyanate, 2,6-toluene diisocyanate, 1,2-naphthalene diisocyanate, 1,5-naphthalene diisocyanate, dianisidine diisocyanate, 3,3'-dimethyl-4,4'-biphenylenediisocyanate and/or pMDI that has been reacted with a hydroxyl-functional compound such as a polyol.

[0042] Even more preferably, the isocyanate-based prepolymer is pMDI, 4,4'-MDI and/or 2,4'-MDI, still more preferably, pMDI.

[0043] Preferably, the epoxy-based prepolymer is an epoxide containing compound or a reaction product between an epoxy compound and, for example, a polypeptide, a polyol, an amine based polyol, an amine containing compound, a hydroxy containing compound or a combination thereof.

[0044] More preferably, the epoxy-based prepolymer is a glycidyl ether-type epoxy resin derived from a polyphenol by reaction with epichlorohydrin such as diglycidyl ether of bisphenol-A, diglycidyl ether of bisphenol-A alkoxylate, diglycidyl ether of 2,6,2',6'-tetrachloro bisphenol A or diglycidyl ether of bisphenol-F (DGEF-F), an epoxy novolac resin such as epoxyphenol novolac or epoxycresol novolac, an epoxidized polyunsaturated vegetable oil such as epoxidized soy oil or epoxidized linseed oil, N,N,N',N'-tetraglycidyl diphenylmethane diamine (such as the 4,4' isomer), a partially or fully saturated (hydrogenated) analog thereof such as DGEF-H (which is the fully hydrogenated (ring saturated) aliphatic analog of DGEF-A), or a combination thereof.

[0045] Even more preferably, the epoxy-based prepolymer is diglycidyl ether of bisphenol-A, diglycidyl ether of bisphenol-A alkoxylate, epoxyphenol novolac, epoxycresol novolac, epoxidized soy oil and/or epoxidized linseed oil.

[0046] Preferably, the latex prepolymer is polyisoprene, polymethylmethacrylate, polymethacrylate, polyvinylchloride, polyvinylacetate, poly(ethylene-vinyl acetate), poly(styrene-butadiene), or a copolymer thereof.

[0047] More preferably, the latex prepolymer is polymethylmethacrylate, polymethacrylate, polyvinylchloride, polyvinylacetate, poly(ethylene-vinyl acetate), or a copolymer thereof.

[0048] Even more preferably, the latex prepolymer is polymethylmethacrylate, polymethacrylate, polyvinylchloride, polyvinylacetate, or a copolymer thereof.

[0049] Advantageously, the dry plant meal is ground before, simultaneously and/or after the heat treatment, preferably, before the heat treatment.

[0050] When the dry plant meal is ground simultaneously with the heat treatment, the grinding and heating steps are advantageously carried out in an attrition mill, a pin mill or an attrition mill wherein hot air is injected.

[0051] In a second aspect, the invention relates to a process for preparing an adhesive composition comprising the steps of: [0052] providing a dry plant meal heat treated at a temperature of at least 100° C., and [0053] mixing the heat treated dry plant meal with water and a reactive prepolymer to provide an adhesive composition, the reactive prepolymer not being soluble in water.

[0054] The above-mentioned definitions and preferred embodiments also apply to this process according to a second aspect of the invention, and in particular, those relating to the dry plant meal and the reactive prepolymer.

[0055] The heat treatment according to the invention may be static or dynamic, for example by using an oven, a pan dryer (such as a Guedu® dryer) or a reactor comprising an endless screw warmed with hot air, a thermal fluid or electrical heating (such as the reactors described in Example 15), in particular, an oven or a pan dryer.

[0056] Preferably, the heat treatment is performed at a temperature of at least 115° C., more preferably at least 135° C., even more preferably, at least 150° C.

[0057] In particular, contrary to a heat treatment performed in liquid solution (e.g. aqueous solution), the use of a dry plant meal allows conducting the heat treatment at a temperature much higher than in an aqueous solution under atmospheric pressure, such as 115° C. or 150° C.

[0058] Advantageously, the maximum temperature for the heat treatment is the ignition temperature of the plant meal, which depends on the plant meal and type of treatment (in cloud or layer). For example, the ignition temperature of rapeseed meal after 5 minutes of heat treatment is 215° C. when rapeseed meal is in “thin” layer (for example, spread out on a plate), and 306° C. when rapeseed meal is in “cloud” (i.e. in suspension in the air).

[0059] Advantageously, the dry plant meal comprises at least 15 wt % of proteins, on the total weight of the dry plant meal.

[0060] Preferably, the dry plant meal comprises at least 25 wt % of proteins, more preferably, at least 30 wt % of proteins, on the total weight of the dry plant meal.

[0061] Advantageously, the dry plant meal and/or the heat treated dry plant meal has a granulometry d50 of at least 20 µm.

[0062] Preferably, the dry plant meal and/or the heat treated dry plant meal has a granulometry d50 of at least 100 µm.

[0063] Preferably, the dry plant meal and/or the heat treated dry plant meal has a granulometry d50 comprised between 20 µm and 500 µm, more preferably, between 30 µm and 300 µm, even more preferably, between 100 µm and 250 µm.

[0064] The granulometry d50 is well known by the skilled person as the maximum size of 50% of the smallest particles in weight, and can be measured with a granulometer, for example according to the method described in Example 1 “Particle size determination”.

[0065] It will be noted that in the context of the present application, and unless otherwise stipulated, the ranges of values indicated are understood to be inclusive.

[0066] Advantageously, the dry plant meal is submitted to a heat treatment during at least 5 minutes.

[0067] Preferably, the dry plant meal is submitted to a heat treatment during at least 15 minutes, more preferably, at least 30 minutes.

[0068] Advantageously, the dry plant meal is submitted to a heat treatment during about 5 min at about 200° C., during about 15 min at about 175° C., during about 30 min at about 150° C. or during about 45 min at about 120° C., preferably, during about 30 min at about 150° C.

[0069] By “about” a value, it is meant said value plus or minus 10%.

[0070] According to a particular embodiment, the dry plant meal is dry rapeseed meal subjected to a heat treatment during about 5 min at about 200° C., during about 15 min at about 150° C. or during about 15 min at about 120° C., preferably, during about 15 min at about 150° C.

[0071] According to another particular embodiment, the dry plant meal is dry sunflower seed meal subjected to a heat treatment during about 5 min at about 200° C., during about 15 min at about 175° C. or during about 30 min at about 150° C., preferably, during about 30 min at about 150° C.

[0072] Advantageously, the heat treated dry plant meal has a dry content greater than 96 wt %, preferably, greater than 98 wt %, more preferably, comprised between 99 wt % and 100 wt %. The dry content can be determined according to the first method set forth in Example 1.

[0073] Advantageously, the weight ratio of the reactive prepolymer/heat treated dry plant meal (dry/dry) is comprised between 0.2 and 8, preferably, between 0.5 and 6, more preferably, between 1.0 and 5, even more preferably, between 2 and 4.

[0074] As mentioned above, the adhesive composition further comprises water. Preferably, the heat treated dry plant meal represents at least 20 wt % of the total weight of the heat treated dry plant meal and water in the adhesive composition, more preferably, at least 25 wt %, even more preferably, at least 27 wt %, the heat treated dry plant meal weight being in dry weight.

Advantageously, the heat treated dry plant meal represents between 20 wt % and 35 wt % of the total weight of the heat treated dry plant meal and water in the adhesive composition, preferably, between 25 wt % and 35 wt %, more preferably, between 27 wt % and 35 wt %, even more preferably, between 27 wt % and 30 wt %, the heat treated dry plant meal weight being in dry weight.

[0075] In a preferred embodiment, the heat treated dry plant meal is added to the adhesive composition under the form of a heat treated dry plant meal dispersion in water. In this preferred embodiment, the process according to the invention further comprises a step of mixing the heat treated dry plant meal with water to provide a heat treated dry plant meal dispersion.

[0076] Advantageously, the heat treated dry plant meal dispersion has a solid content of at least 20 wt %, preferably, at least 25 wt %, more preferably, at least 27 wt %.

[0077] By “solid content” of a plant meal dispersion, it is meant the percentage of dry solids contained in said dispersion. For example, if a dispersion is prepared by mixing plant meal and water, the residual water already contained the plant meal is taken in consideration for calculating the solid content of the dispersion.

[0078] Advantageously, the heat treated dry plant meal dispersion has a solid content comprised between 20 wt % and 35 wt %, preferably, comprised between 25 wt % and 35 wt %, more preferably, comprised between 27 wt % and 35 wt %, even more preferably, comprised between 27 wt % and 30 wt %.

[0079] Advantageously, the heat treated dry plant meal dispersion has a viscosity at 20° C. of less than 1000 mPa.Math.s, preferably, of less than 500 mPa.Math.s, more preferably, of less than 400 mPa.Math.s, still more preferably, of less than 300 mPa.Math.s, even more preferably, of less than 200 mPa.Math.s.

[0080] The skilled person knows how to measure viscosity. Preferably, the viscosity can be

measured as described in Example 1.

[0081] Furthermore, all the viscosity measurements are done less than 15 min after stopping mixing a composition comprising the heat treated dry plant meal and water.

[0082] According to a preferred embodiment, the heat treated dry plant meal dispersion has the above described viscosity and a solid content comprised between 25 wt % and 35 wt %, preferably, comprised between 27 wt % and 35 wt %, more preferably, comprised between 27 wt % and 30 wt %.

[0083] Advantageously, the adhesive composition has a viscosity at 20° C. of less than 1000 mPa.Math.s, preferably, of less than 500 mPa.Math.s, more preferably, of less than 400 mPa.Math.s, still more preferably, of less than 300 mPa.Math.s, even more preferably, of less than 200 mPa.Math.s.

[0084] According to a preferred embodiment, the adhesive composition has the above described viscosity and comprises between 25 wt % and 35 wt % of heat treated dry plant meal, preferably, between 27 wt % and 35 wt %, more preferably, between 27 wt % and 30 wt %, the percentages being indicated with respect to the total weight of the heat treated dry plant meal and water in the adhesive composition, and the heat treated dry plant meal weight being in dry weight.

[0085] The heat treated dry plant meal dispersion and/or the adhesive composition may also comprise one or more additives.

[0086] Exemplary additives include clays, in particular, an intercalated clay, partially exfoliated clay, exfoliated clay, for example montmorillonite, cellulose nanoparticles, catalysts, tacking agents, extenders, fillers, viscosifying agents, surfactants, adhesion promoters, antioxidants, antifoaming agents, antimicrobial agents, antibacterial agents, fungicides, pigments, inorganic particulates (e.g., titanium dioxide, yellow iron oxide, red iron oxide, black iron oxide, zinc oxide, aluminum oxide, aluminum trihydrate, calcium carbonate), gelling agents, aereosolizing agents, cross-linking agents, agents that improve moisture resistance, pH modulators, composite-release promoters, formaldehyde scavenging agents, fire retardants, wetting agents, and wood preservatives.

[0087] The additive may be a water-dispersible additive or a water-soluble additive. Water-soluble additives include hydroxyl-functional or amine-functional compounds (such as glycerin, propylene glycol, polypropylene glycol, polyethylene glycol, trimethylol propane and its adducts, phenols, polyphenols, etc.). One benefit of using glycerin and various low-viscosity polyols is that they allow less water to be used in the adhesive composition. The benefit comes in addition to heat treated dry plant meal in an adhesive composition. It allows to further increase the solid content versus unheated treated dry plant meal.

[0088] Reducing the amount of water, while retaining a low-viscosity adhesive composition, desirably reduces the risk that the composite formed therefrom is damaged by steam generated during formation of the composite at high temperature.

[0089] The additive may also be a non-volatile (e.g., having a boiling point of greater than about 180° C. at atmospheric pressure), inert viscosity-reducing diluent.

[0090] The additive may be an agent that improves moisture-resistance, a composite-release promoter (such as a composite-release promoter selected from the group consisting of a C.sub.10-25 alkanolic acid, a salt of a C.sub.10-25 alkanolic acid, a C.sub.10-25 alkenolic acid, a salt of an C.sub.10-25 alkenolic acid, and a silicone).

[0091] In particular, the additive may be a fire retardant, wood preservative, antimicrobial agent, antibacterial agent or fungicide, any of which may be in the form of nanoparticles.

[0092] Advantageously, if the adhesive comprises one or more additives, each additive present in the adhesive composition is independently present in an amount ranging from 0.001% (w/w) to about 20% (w/w), from 0.01% (w/w) to about 15% (w/w) from 0.1% (w/w) to about 10% (w/w), from 0.5% (w/w) to about 3% (w/w), from 1% (w/w) to about 20% (w/w), from 1% (w/w) to about 10% (w/w), from 1% (w/w) to about 5% (w/w), from 1% (w/w) to about 3% (w/w) or from 5%

(w/w) to about 10% (w/w). In certain embodiments, such as where the additive is a fire retardant, the additive may be present in the adhesive composition in an amount ranging from about 1% (w/w) to about 40% (w/w), from about 10% (w/w) to about 40% (w/w), from about 20% (w/w) to about 40% (w/w), or from about 25% (w/w) to about 35% (w/w).

[0093] Exemplary classes of additives are described in more detail below.

Intercalated Clay

[0094] Intercalated clays can be obtained from commercial sources or prepared by exposing a clay to an intercalating agent. Exemplary types of clay that may be converted to intercalated form include, for example, smectite clays, illite clays, chlorite clays, layered polysilicates, synthetic clays and phyllosilicates. Exemplary specific clays that may be converted to intercalated form include, for example, montmorillonite (e.g., sodium montmorillonite, magnesium montmorillonite and calcium montmorillonite), beidellite, pyrophyllite, talc, vermiculite, sobockite, stevensite, svinfordite, sauconite, saponite, volkonskoite, hectorite, nontronite, kaolinite, dickite, nacrite, halloysite, hisingerite, rectorite, tarosovite, ledikite, amesite, baileychlore, chamosite, clinochlore, kaemmererite, cookeite, corundophilite, daphnite, delessite, gonyerite, nimate, odinite, orthochamosite, penninite, pannantite, rhipidolite, prochlorite, sudoite, thuringite, kanemite, makatite, ilerite, octosilicate, magadiite and kenyaite. In certain embodiments, the clay converted to intercalated form is montmorillonite.

[0095] Exemplary intercalating agents include, for example, quaternary amine compounds (such as a tetra-alkylammonium salt), polymers (e.g., a polycaprolactone, maleated polyethylene, or maleated polypropylene) an acrylic monomer, phosphonium compounds, arsonium compounds, stibonium compounds, oxonium compounds, sulfonium compounds, polypropylene, fatty acid esters of pentaerythritol, a steroyl citric acid ester and alcohols (such as aliphatic alcohols, aromatic alcohols (e.g., phenols), aryl substituted aliphatic alcohols, alkyl substituted aromatic alcohols, and polyhydric alcohols).

[0096] Intercalated clays can be characterized by, for example, the following physical properties: interlayer spacing, d-spacings, clay particle size, particle size distribution, peak degradation temperature, and thickness of layers. Exemplary physical property features for intercalated clays contemplated to be amenable for use in the present invention include, for example, one or more of the following: (i) an intercalated clay having an interlayer spacing of about 0.5 Å to about 100 Å (or about 1 Å to about 20 Å), (ii) a mean particle size of about 1 μm to about 150 μm (or about 20 μm to about 100 μm), (iii) a particle size distribution where about 90 percent to about 50 percent of the intercalated clay particles have a particle size of from about 20 μm to about 100 μm (or about 85 percent to about 65 percent of the intercalated clay particles have a particle size of about 20 μm to about 100 μm), (iv) a peak degradation temperature of about 200° C. to about 600° C. (or from about 300° C. to about 500° C.), and/or (v) layers in the intercalated clay have a thickness of about 0.5 Å to about 100 Å (or about 5 Å to about 50 Å).

[0097] In certain other embodiments, the intercalated clay is intercalated montmorillonite having a particle size of less than about 500 nm, or less than about 100 nm. In certain other embodiments, the intercalated clay is intercalated montmorillonite having a particle size of about 60 nm to about 400 nm.

[0098] The clay (e.g., an intercalated clay) may be surface treated with an organic compound, such as a hydrophobic organic compound or hydrophilic organic compound, in order to promote dispersion of the clay in a formulation, such as an adhesive composition described herein. Surface treatment methods and compositions are described in the literature and are contemplated to be amenable for use in the present invention.

[0099] Different intercalated clays may impart different performance properties to the adhesive composition. Accordingly, in certain embodiments, the intercalated clay is an intercalated smectite. In certain other embodiments, intercalated clay is a smectite that has been intercalated with a quaternary ammonium compound. In certain other embodiments, the intercalated clay is an

intercalated montmorillonite. In yet other embodiments, the intercalated clay is montmorillonite intercalated with a dimethyl-di(C.sub.14-18) alkyl ammonium salt.

Exfoliated Clay & Partially Exfoliated Clay

[0100] Exfoliated clay or a partially exfoliated clay can be prepared by exposing an intercalated clay to exfoliation conditions using procedures described in the literature. One procedure for preparing a partially exfoliated clay is to subject an intercalated clay to high shear mixing and/or sonication (e.g., using ultrasound) until the intercalated clay has partially exfoliated. The procedure may be performed by placing the intercalated clay (e.g., quaternary amine intercalated montmorillonite) in a hydrophobic liquid medium (such as mineral oil, soy oil, castor oil, silicone oil, a terpene (e.g., limonene), plant oil alkyl esters (e.g., soy methyl ester and canola methyl ester), mixtures thereof (e.g., a mixture of a silicone oil and limonene), etc.) to form a mixture, and then subjecting the mixture to high shear mixing and/or ultrasound until the intercalated clay has partially exfoliated. Partial exfoliation occurs when clay platelets separate from the intercalated clay particles. The intercalated clay may be added to the adhesive composition, and the adhesive composition is subjected to exfoliation conditions to generate the partially exfoliated clay in situ.

[0101] An exfoliated clay can be prepared by exposing an intercalated clay to high shear mixing and/or sonication (e.g., using ultrasound) until substantially all (e.g., greater than 90% w/w, 95% w/w, or 98% w/w) the intercalated clay has exfoliated. The exfoliation procedure can be performed by placing the intercalated clay (e.g., quaternary amine intercalated montmorillonite) in a hydrophobic liquid medium (such as mineral oil, soy oil, castor oil, silicone oil, a terpene (e.g., limonene), plant oil alkyl esters (e.g., soy methyl ester and canola methyl ester), mixtures thereof (e.g., a mixture of a silicone oil and limonene), etc.) to form a mixture, and then subjecting the mixture to high shear mixing and/or sonication (e.g., using ultrasound) until substantially all (e.g., greater than 90% w/w, 95% w/w, or 98% w/w) the intercalated clay has exfoliated. Alternatively, the intercalated clay may be added to the adhesive composition, and the adhesive composition is subjected to exfoliation conditions to generate the exfoliated clay in situ. Alternatively, a clay (such as sodium montmorillonite) may be added to an adhesive composition, together with a quaternary ammonium compound, and optionally together with a satisfactory oil carrier (e.g., one that has the ability to solvate the quaternary compound), and the resulting adhesive composition is subjected to conditions to intercalate the clay and to generate the exfoliated clay or partially exfoliated clay in situ. In addition, if so desired, the quaternary ammonium compound can be pre-dissolved in the oil carrier before it is added to the adhesive composition together with a clay.

[0102] Exemplary partially exfoliated clays contemplated to be amenable for use in present invention include partially exfoliated forms of smectite clay, illite clay, chlorite clay, layered polysilicates, synthetic clay and phyllosilicates. Exemplary specific partially exfoliated clays contemplated to be amenable for use in present invention include partially exfoliated forms of, for example, montmorillonite (e.g., sodium montmorillonite, magnesium montmorillonite, and calcium montmorillonite), beidellite, pyrophyllite, talc, vermiculite, sobockite, stevensite, svinfordite, sauconite, saponite, volkonskoite, hectorite, nontronite, kaolinite, dickite, nacrite, halloysite, hisingerite, rectorite, tarosovite, ledikite, amesite, baileychlore, chamosite, clinochlore, kaemmererite, cookeite, corundophilite, daphnite, delessite, gonyerite, nimite, odinite, orthochamosite, penninite, pannantite, rhipidolite, prochlorite, sudoite, thuringite, kanemite, makatite, ilerite, octosilicate, magadiite and kenyaite. In certain embodiments, the partially exfoliated clay is partially exfoliated clay montmorillonite.

[0103] A partially exfoliated clay can be characterized by, for example, the amount of clay particles that are in the form of platelets. In certain embodiments, about 0.1% w/w to about 40% w/w, about 0.1% w/w to about 20% w/w, about 0.1% w/w to about 10% w/w, about 0.1% w/w to about 5% w/w, about 5% w/w to about 20% w/w of the clay particles are in the form of platelets. In certain embodiments, about 0.1% w/w to about 40% w/w of the clay particles are in the form of platelets having a size of about 1 Å to about 50 Å, about 30 Å to about 50 Å or about 5 Å to about 20 Å.

[0104] Exemplary exfoliated clays contemplated to be amenable for use in present invention include exfoliated forms of smectite clay, illite clay, chlorite clay, layered polysilicates, synthetic clay and phyllosilicates. Exemplary specific exfoliated clays contemplated to be amenable for use in present invention include exfoliated forms of, for example, montmorillonite (e.g., sodium montmorillonite, magnesium montmorillonite, and calcium montmorillonite), beidellite, pyrophyllite, talc, vermiculite, sobockite, stevensite, svinfordite, sauconite, saponite, volkonskoite, hectorite, nontronite, kaolinite, dickite, nacrite, halloysite, hisingerite, rectorite, tarosovite, ledikite, amesite, baileychlore, chamosite, clinochlore, kaemmererite, cookeite, corundophilite, daphnite, delessite, gonyerite, nimate, odinite, orthochamosite, penninite, pannantite, rhipidolite, prochlore, sudoite, thuringite, kanemite, makatite, ilerite, octosilicate, magadiite and kenyaite. In certain embodiments, the exfoliated clay is an exfoliated smectite. In certain embodiments, the exfoliated clay is exfoliated montmorillonite.

[0105] An exfoliated clay can be characterized by, for example, the size of platelets and the aspect ratio of platelets. In certain embodiments, the size of the platelets is about 1 Å to about 50 Å, about 30 Å to about 50 Å or about 5 Å to about 20 Å. In certain embodiments, the aspect ratio of the platelets is about 100 to about 10000, about 100 to about 5000 or about 200 to about 2000. In certain other embodiments, the exfoliated clay has a mean particle size of less than about 500 nm, less than 100 nm or less than 25 nm. In certain other embodiments, the exfoliated clay has a mean particle size of from about 60 nm to about 400 nm, about 50 nm to about 300 nm, about 40 nm to about 200 nm or about 20 nm to about 150 nm.

[0106] In certain other embodiments, a partially exfoliated clay is formed by exposing a clay to an effective amount of a protein component (e.g., dry plant meal or heat treated dry plant meal) to form a mixture and subjecting the mixture to exfoliation conditions, such as high shear mixing and/or sonication. In certain other embodiments, an exfoliated clay is formed by exposing a clay to an effective amount of protein component (e.g., dry plant meal or heat treated dry plant meal) to form a mixture and subjecting the mixture to exfoliation conditions, such as high shear mixing and/or sonication.

Cellulose Nanoparticles

[0107] Cellulose nanoparticles can be added to the adhesive composition to achieve certain performance properties, such as to provide an adhesive composition with increased toughness and/or bond strength. Cellulose nanoparticles can be obtained from commercial sources or isolated from plant-based fibers by acid-hydrolysis. Cellulose nanoparticles can be characterized by, for example, the size of the nanoparticle, the cross-sectional shape of the nanoparticle, and the cross-sectional length and aspect ratio of the nanoparticle.

[0108] Accordingly, in certain embodiments, the cellulose nanoparticle has a size of from about 1 nm to about 2000 nm, about 10 nm to about 1000 nm, about 10 nm to about 500 nm or about 10 nm to about 200 nm. In certain embodiments, the cross-sectional shape of the nanoparticle may be triangular, square, pentagonal, hexagonal, octagonal, circular, or oval. In certain other embodiments, the average cross-sectional length of the cellulose nanoparticle is about 0.1 nm to about 100 nm, or about 1 nm to about 10 nm.

[0109] One type of cellulose nanoparticles that may provide certain advantages are cellulose nanofibers. Exemplary cellulose nanofibers are described in, for example, U.S. Patent Application Publication Nos. 2010/0233481, 2010/0240806 and 2010/0282422.

Catalyst

[0110] A catalyst may be added to the adhesive composition to facilitate polymerization. Exemplary catalysts include, for example, a primary amine, a secondary amine, a tertiary amine, an organometallic compound, or a combination thereof. Exemplary primary amines include, for example, methylamine, ethylamine, propylamine, cyclohexylamine, and benzylamine. Exemplary secondary amines include, for example, dimethylamine, diethylamine, and diisopropylamine. Exemplary tertiary amines include, for example, diazabicyclooctane (Dabco), triethylamine,

dimethyl benzylamine, bis-dimethylaminoethyl ether, tetramethyl guanidine, bis-dimethylaminomethyl phenol, 2,2'-dimorpholinodiethyl ether, 2-(2-dimethylaminoethoxy)-ethanol, 2-dimethylaminoethyl-3-dimethylaminopropyl ether, bis-(2-diaminoethyl)-ether, N,N-dimethyl piperazine, N-(2-hydroxyethoxyethyl)-2-azanorbornane, Tacat DP-914 (Texaco Chemical), Jeffcat®, N,N,N,N-tetramethyl butane-1,3-diamine, N,N,N,N-tetramethyl propane-1,3-diamine, N,N,N,N-tetramethyl hexane-1,6-diamine, 2,2'-dimorpholinodiethyl ether (DMDEE), or a mixture thereof. Exemplary organometallic compounds include, for example, di-n-octyl tin mercaptide, dibutyl tin maleate, diacetate, dilaurate, dichloride, bis-dodecyl mercaptide, tin (II) acetate, ethyl hexoate and diethyl hexoate, Fe.sup.3+ 2,4-pentanedionate (FeAcAc) or lead phenyl ethyl dithiocarbamate.

[0111] In certain other embodiments, the catalyst is a transition metal acetylacetonate, e.g., an acetylacetonate compound comprising iron, copper, or nickel. In certain embodiments, the transition metal acetylacetonate comprises a tertiary amine, e.g., 2,2'-dimorpholino diethyl ether.

[0112] The amount of catalyst used in the adhesive composition can be varied in order to optimize the features of the adhesive composition. In certain embodiments, the catalyst is present in less than 1% (wt/wt), 0.5% (wt/wt) or 0.1% (wt/wt) of the adhesive composition. In certain other embodiments, the catalyst is present in a range from 0.001% (wt/wt) to 0.75% (wt/wt), 0.001% (wt/wt) to 0.01% (wt/wt), 0.01% (wt/wt) to 0.05% (wt/wt) or 0.05% (wt/wt) to 0.5% (wt/wt) of the adhesive composition.

Tacking Agent

[0113] Exemplary tacking agents include, for example, glycerin, corn syrup, soy oil, a poly(C.sub.2-C.sub.6) alkylene, mineral oil, an ethylene/propylene/styrene copolymer, a butylene/ethylene/styrene copolymer, or a mixture of one or more of the foregoing. Other exemplary tacking agents are copolymers that have a low glass transition temperature (T_g) (e.g., a latex-based, acrylic copolymer with a T_g of less than about 0° C., and preferably less than about -20° C.). In certain embodiments, the additive is polybutene. In certain embodiments, the polybutene has a weight average molecular weight of from about 200 g/mol to about 20000 g/mol, from about 200 g/mol to about 10000 g/mol, from about 200 g/mol to about 5000 g/mol, from about 200 g/mol to about 2000 g/mol, from about 200 g/mol to about 1000 g/mol, from about 500 g/mol to about 2000 g/mol, or from about 500 g/mol to about 1000 g/mol. Other tacking agents include a solid selected from the group consisting of a terpene resin, a rosin ester derivative, and a hydrocarbon-based derivative. When the tacking agent is a solid, the tacking agent may optionally be pre-dissolved in an oil-phase of the adhesive composition (e.g., in PMDI). Alternatively, the solid tacking agent can be pre-melted and dispersed in water by means of the heat treated dry plant meal, or the solid tacking agent can be ground and dispersed as fine particulates directly into the adhesive composition.

Extender

[0114] Exemplary extenders include, for example, inert extenders or active extenders. In certain embodiments, the inert extender is vegetable particulate matter, limonene, vegetable oil, mineral oil, dibasic esters, propylene carbonate, non-reactive modified aromatic petroleum hydrocarbons, soy oil, castor oil, and in general any non-active hydrogen containing liquid that can be incorporated into an isocyanate based adhesive. Another inert extender is any non-active hydrogen containing solid that is soluble, e.g., soluble in oil or soluble in water. The active extender can be a pyrrolidone monomer or polymers, an oxizolidone monomer or polymers, an epoxidized oil, or an unsaturated oil, such as linseed oil. Another active extender is a vinyl monomer or mixture of vinyl monomers.

Surfactants & Adhesion Promoters

[0115] Exemplary surfactants include, for example, monomeric types, polymeric types, or mixtures thereof. Exemplary adhesion promoters include, for example, organosilanes and titanates.

Antimicrobial Agent

[0116] Antimicrobial agents known in the art that do not substantially react with the reactive prepolymer are contemplated for use in the adhesive compositions and composites described herein. One exemplary antimicrobial agent is polyalkylene glycol polymers, such as polypropylene glycol.

Crosslinking Agent

[0117] In other embodiments, the additive can be a crosslinking agent, for example, a crosslinking agent that can be used to bond lignocellulosic material to glass. Exemplary crosslinking agents include an organosilane, such as dimethyldichlorosilane (DMDCS), alkyltrichlorosilane, methyltrichlorosilane (MTCS), N-(2-aminoethyl)-3-aminopropyl trimethoxysilane (AAPS), or a combination thereof. In other embodiments the heat treated dry plant meal is combined with an organosilane to form an adhesive for bonding one or more substrates together in any combination, said substrates including glass, paper, wood, ceramic, steel, aluminum, copper, brass, etc. The term “organosilane” refers to any group of molecules including monomers, hydrolyzed monomers, hydrolyzed dimers, oligomers, and condensation products of a trialkoxysilane having a general formula: $(\text{RO})_3\text{Si}-\text{R}'$ where R is preferably a propyl, ethyl, methyl, isopropyl, butyl, isobutyl, sec-butyl, t-butyl or acetyl group, and R' is an organofunctional group where the functionality may include an aminopropyl group, an aminoethylaminopropyl group, an alkyl group, a vinyl group, a phenyl group, a mercapto group, a styrylamino group, a methacryloxypropyl group, a glycidoxy group, an isocyanate group, or others.

[0118] Similarly, a bis-trialkoxysilane having the general formula $(\text{RO})_3\text{Si}-\text{R}-\text{Si}(\text{OR})_3$ can also be employed as an “organosilane” either alone or in combination with a trialkoxysilane, where R is preferably a propyl, ethyl, methyl, isopropyl, butyl, isobutyl, sec-butyl, t-butyl or acetyl group, and R is a bridging organofunctional residue which may contain functionality selected from the group consisting of amino groups, alkyl groups, vinyl groups, phenyl groups, mercapto groups, and others. Similarly, a tetraalkoxysilane having the general formula $(\text{RO})_4\text{Si}$ can also be employed as an “organosilane” either alone or in combination with a trialkoxysilane or a bis-trialkoxysilane, where R is preferably a propyl, ethyl, methyl, isopropyl, butyl, isobutyl, sec-butyl, t-butyl or acetyl group.

Agent that Improves Moisture-Resistance

[0119] Agents that improve moisture-resistance refer to those materials that, when added to adhesive compositions described herein, improve the ability of a wood composite formed from the adhesive to be resistant to water, i.e., not absorb water. Exemplary types of agents that improve moisture resistance include fluorinated polyol compounds, silicones, siloxanes (including functionalized siloxane polymers, such as hydroxy-terminated siloxane polymers or hydroxyl alkyl siloxane polymers), polyolefin polymers, wax (e.g., fatty acids (such as an alkyl carboxylic acid), salts of a fatty acid (e.g., an alkali metal salt of an alkyl carboxylic acid), esters of a fatty acid (e.g., an alkyl ester of a carboxylic acid, an aryl ester of a carboxylic acid, an alkyl ester of an alkanolic acid, or an aryl ester of an alkanolic acid), fatty alcohols, mixtures of hydrophobic hydrocarbons, water-based emulsions containing hydrophobic hydrocarbons dispersed therein, a hydrocarbon wax, a fluoroalkylphosphate wax, a fluorinated hydrocarbon wax, and a fluoroalkyl functionalized wax), and hydrophobic oils. Another agent that improves moisture-resistance is a fluorinated silicone. When an agent that improves moisture-resistance is present in an adhesive composition, it is desirably present in an amount effective to increase moisture resistance (e.g., an increase in moisture resistance of at least about 2%, 5%, 10%, or 20% compared to the moisture resistance of a composite formed from an adhesive composition lacking the agent that improves moisture-resistance).

[0120] Agents that improve moisture-resistance may be present in the final composite at a weight percent in the range of about 0.01% (w/w) to about 5% (w/w), about 0.01% (w/w) to about 2% (w/w), about 0.01% (w/w) to about 1% (w/w), about 0.01% (w/w) to about 0.5%, about 0.1% (w/w) to about 2% (w/w), (w/w), about 0.1% (w/w) to about 1% (w/w) or about 0.5% (w/w) to

about 1% (w/w).

[0121] One exemplary fluorinated polyol compound is FluoroLink D-10 fluorinated polyol that is commercially available from Solvay Solexis, Inc. Exemplary silicones include Dow Corning FS-1265 Fluid, 300 cST (Trifluoropropyl Methicone) from Dow Corning), and T-Sil-6011 SE Emulsion (60% Solids), from Siovation, Inc. which is a emulsion containing 60% w/w silicones. The silicone may be organically modified, such as C.sub.20-24 Alkyl Methicone, C.sub.24-28 Alkyl Methicone, C.sub.30-45 Alkyl Methicone, Stearyl Dimethicone, Biphenyl Dimethicone, Stearoxy Dimethicone, C.sub.20-24 Alkyl Dimethicone, or C.sub.24-28 Alkyl Dimethicone.

[0122] Exemplary types of functionalized siloxane polymers include (1) a hydroxy-terminated siloxane polymer such as hydroxy-terminated polydimethylsiloxane (e.g., T-Sil-80, a linear hydroxy terminated polydimethylsiloxane from Siovation, Inc.), (2) a hydroxyl alkyl polydimethylsiloxane (e.g., Tego Protect-5000 functionalized silicone fluid commercially available from Evonik Tego Chemie GmbH), and (3) a fluorinated siloxane.

[0123] Exemplary waxes include Hexion EW-58H; BE Square 165 Amber Petroleum Macrocrystalline Wax commercially available from Baker Hughes, Inc., which is a mixture comprising paraffinic, isoparaffinic, and naphthenic hydrocarbons; Masurf FS 115 Emulsion (a fluoroalkylphosphate wax dispersion in water—28% Solids) commercially available from Mason Chemical Company; carnauba wax; candelilla wax; japan wax; beeswax; rice bran wax; montan wax; paraffin; ceresin; lanolin; ozokerita; slack wax (which is semi-refined wax having an oil content up to about 30 mass percent, and often produced by chilling and solvent filter-pressing wax distillate); polyethylene wax; a fatty acid or salt thereof (e.g., C.sub.10-25 alkanolic acid, a salt of a C.sub.10-25 alkanolic acid, a C.sub.10-25 alkenolic acid, a salt of an C.sub.10-25 alkenolic acid; such as stearic acid, zinc stearate, or lauric acid), a fatty ester (e.g., an ester of an C.sub.10-25 alkanolic acid or C.sub.10-25 alkenolic acid); or fatty alcohol (e.g., C.sub.10-25 hydroxy alkane or C.sub.10-25 hydroxy alkene). Preferably, the wax is paraffin, carnauba wax and/or beeswax, more preferably, paraffin.

[0124] Exemplary hydrophobic polymers include a polyolefin (e.g., polyethylene, polypropylene, polybutylene, polystyrene, copolymers of the foregoing, polyethylene/polyvinyl acetate copolymer, and polyethylene/polyacrylic acid copolymer).

[0125] Exemplary hydrophobic oils include soy lecithin, castor oil, and a fluorinated hydrocarbon liquid.

[0126] Another agent that improves moisture resistance is a mixture of a silicone and a terpene compound. An exemplary silicone is Tego Protect-5000 functionalized silicone fluid sold by Evonik Tego Chemie GmbH. Exemplary terpene compounds contemplated for use include terpene compounds that are a solid at room temperature, a liquid at room temperature, and/or have a molecular weight of less than about 2000 g/mol, about 1000 g/mol, about 500 g/mol, or about 200 g/mol. In certain embodiments, the terpene compound is limonene. In certain embodiments, the agent that improves moisture resistance is a mixture of Tego Protect-5000 functionalized silicone fluid and limonene.

[0127] In certain embodiments, the agent that improves moisture-resistance is a polymer agent that improves moisture-resistance, a wax agent that improves moisture-resistance, or a mixture thereof. In certain other embodiments, the agent that improves moisture-resistance is a silicone, a siloxane, a fluorinated polyol, a fluoroalkyl phosphate ester, a fluoroalkyl carboxylic ester, a salt of a fluoroalkanoic acid, a wax that improves moisture-resistance, or a mixture thereof. In certain other embodiments, the agent that improves moisture-resistance is a wax that improves moisture-resistance, such as a mixture of hydrophobic hydrocarbons, water-based emulsions containing hydrophobic hydrocarbons dispersed therein, a fluoroalkylphosphate wax, a fluorinated hydrocarbon wax, or a fluoroalkyl functionalized wax. In certain other embodiments, the agent that improves moisture-resistance is a silicone, a siloxane, a fluorinated polyol, a fluoroalkyl phosphate ester, or a fluoroalkyl carboxylic ester. In certain other embodiments, the agent that improves

moisture-resistance is a silicone, a siloxane, a fluorinated polyol, a fluoroalkyl phosphate ester, a fluoroalkyl carboxylic ester, a salt of a fluoroalkanoic acid, or a mixture thereof. In certain other embodiments, the agent that improves moisture-resistance is a silicone, a siloxane, a fluorinated polyol, a fluoroalkyl phosphate ester, a fluoroalkyl carboxylic ester, or a wax that improves moisture-resistance. In certain other embodiments, the agent that improves moisture-resistance is a fluorinated polyol, a silicone, a siloxane, or wax that improves moisture-resistance. In yet other embodiments, the agent that improves moisture-resistance is a mixture comprising hydrophobic hydrocarbons.

[0128] The term “fluoroalkyl phosphate ester” as used herein refers to a compound comprising a phosphate group bonded to at least one fluoroalkyl group, such as represented by $P(O)(OR.^{sup.1})(OR.^{sup.2})._{sub.2}$, wherein $R.^{sup.1}$ is a fluoroalkyl group, and $R.^{sup.2}$ represents independently for each occurrence hydrogen, alkyl, fluoroalkyl, aryl, aralkyl, heteroalkyl, heteroaryl, heteroaralkyl, an alkali metal, ammonium, or a quaternary amine, or two occurrences of $R.^{sup.2}$ are taken together to form an alkaline earth metal.

pH Modulator

[0129] The pH modulator can be an acid or base. In certain embodiments, the pH modulator is an alkali metal hydroxide (e.g., sodium hydroxide or calcium hydroxide) or an alkali metal salt of a carboxylate organic compound (e.g., an alkali metal salt of citrate, such as disodium citrate).

Composite-Release Promoter

[0130] The composite-release promoter acts to facilitate release of the wood composite from the press apparatus used to make the composite. In the absence of a composite-release promoter, certain composites may stick to the press apparatus, making it difficult to separate the composite from the press apparatus. The composite-release promoter solves this problem by facilitating release of the wood composite. Exemplary composite-release promoters include silicones (e.g., silicones described above), fatty acids, a salt of a fatty acid, waxes, and amide compounds. Exemplary fatty acids or salts thereof include a $C.^{sub.10-25}$ alkanolic acid, a salt of a $C.^{sub.10-25}$ alkanolic acid, a $C.^{sub.10-25}$ alkenolic acid, a salt of a $C.^{sub.10-25}$ alkenolic acid; e.g., stearic acid, zinc stearate, lauric acid, oleic acid or a salt thereof (such as an alkali metal salt of oleic acid, such as potassium oleate). It is understood that a mixture of two or more of the aforementioned exemplary composite-release promoters can also be used in the adhesive compositions herein. An exemplary amide compound is N,N' -ethylenebisstearamide. Exemplary waxes include those described above for the agent that improves moisture resistance, and in particular, Hexion EW-58H; E Square 165 Amber Petroleum Macrocrystalline Wax commercially available from Baker Hughes, Inc.; and Masurf FS 115 Emulsion (28% Solids) commercially available from Mason Chemical Company. One additional advantage of the heat treated dry plant meal in the adhesive composition is that it can facilitate dispersion of the composite-release promoter—this feature allows less composite-release promoter to be used in the adhesive composition and final composite product. Reducing the amount of composite-release promoter is advantageous for agents that are relatively more expensive, such as certain silicone composite-release promoters.

[0131] In certain embodiments, the composite-release promoter is a silicone.

[0132] Further, in certain embodiments, a composite-release promoter is present in the final composite at a weight percent in the range of about 0.01% (w/w) to about 5% (w/w), about 0.01% (w/w) to about 2% (w/w) or about 0.01% (w/w) to about 1% (w/w).

Formaldehyde Scavenging Agent

[0133] A variety of formaldehyde scavenging agents are described in the literature and are contemplated to be amenable to the present invention. Different formaldehyde scavenging agents have different reactivity profiles, and a particular formaldehyde scavenging agent (e.g., $H.^{sub.2}NC(O)NH.^{sub.2}$, $Me.^{sub.2}NC(O)NH.^{sub.2}$, or $CH.^{sub.3}CH.^{sub.2}NH.^{sub.2}$) can be selected to optimize the performance properties of the adhesive composition and/or binder composition formed by the adhesive. Accordingly, in certain embodiments, the formaldehyde

scavenging agent has the formula RNH.sub.2 , R.sub.2NH , RC(O)NH.sub.2 , R(H)C(O)NH.sub.2 , $\text{R.sub.2NC(O)NH.sub.2}$, or R(H)C(O)N(H)R , wherein R represents independently for each occurrence H, alkyl, aryl, or aralkyl. In certain embodiments, the formaldehyde scavenging agent has the formula RN(H)C(O)N(H)R , wherein R represents independently for each occurrence H, alkyl, aryl, or aralkyl. In certain other embodiments, the formaldehyde scavenging agent is $\text{H.sub.2NC(O)NH.sub.2}$, $\text{H.sub.2NC(O)N(H)Me}$, MeN(H)C(O)N(H)Me , $\text{H.sub.2NC(O)N(CH.sub.3).sub.2}$, $\text{CH.sub.3C(O)NH.sub.2}$, $\text{CH.sub.3CH.sub.2C(O)NH.sub.2}$, CH.sub.3NH.sub.2 , $\text{CH.sub.3CH.sub.2NH.sub.2}$, $(\text{CH.sub.3}).sub.2\text{NH}$ or $(\text{CH.sub.3CH.sub.2}).sub.2\text{NH}$. In still other embodiments, the formaldehyde scavenging agent is $\text{H.sub.2NC(O)NH.sub.2}$.

[0134] The term “alkyl” as used herein refers to a saturated straight or branched hydrocarbon, such as a straight or branched group of 1-12, 1-10, or 1-6 carbon atoms, referred to herein as C.sub.1-C.sub.12 alkyl, C.sub.1-C.sub.10 alkyl, and C.sub.1-C.sub.6 alkyl, respectively. Exemplary alkyl groups include, but are not limited to, methyl, ethyl, propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, butyl, isobutyl, t-butyl, pentyl, isopentyl, neopentyl, hexyl, heptyl, octyl, etc.

[0135] The term “aryl” as used herein refers to a mono-, bi-, or other multi-carbocyclic, aromatic ring system. Unless specified otherwise, the aromatic ring is optionally substituted at one or more ring positions with substituents selected from alkanoyl, alkoxy, alkyl, alkenyl, alkynyl, amido, amidino, amino, aryl, arylalkyl, azido, carbamate, carbonate, carboxy, cyano, cycloalkyl, ester, ether, formyl, halogen, haloalkyl, heteroaryl, heterocyclyl, hydroxyl, imino, ketone, nitro, phosphate, phosphonato, phosphinato, sulfate, sulfide, sulfonamido, sulfonyl and thiocarbonyl. The term “aryl” also includes polycyclic ring systems having two or more cyclic rings in which two or more carbons are common to two adjoining rings (the rings are “fused rings”) wherein at least one of the rings is aromatic, e.g., the other cyclic rings may be cycloalkyls, cycloalkenyls, cycloalkynyls, and/or aryls.

[0136] Exemplary aryl groups include, but are not limited to, phenyl, tolyl, anthracenyl, fluorenyl, indenyl, azulenyl, and naphthyl, as well as benzo-fused carbocyclic moieties such as 5,6,7,8-tetrahydronaphthyl. In certain embodiments, the aryl group is not substituted, i.e., it is unsubstituted.

[0137] The term “aralkyl” as used herein refers to an aryl group having at least one alkyl substituent, e.g. aryl-alkyl-. Exemplary aralkyl groups include, but are not limited to, arylalkyls having a monocyclic aromatic ring system, wherein the ring comprises 6 carbon atoms. For example, “phenylalkyl” includes phenylC4 alkyl, benzyl, 1-phenylethyl, 2-phenylethyl, etc.

[0138] The amount of formaldehyde scavenging agent in the adhesive formulation can be adjusted to optimize the performance properties of the adhesive composition and/or binder composition formed by the adhesive. In certain embodiments, the mole ratio of formaldehyde scavenging agent to reactive prepolymer is at least about 0.1:1, at least about 0.5:1, at least about 1:1, at least about 2:1, at least about 3:1, at least about 4:1 or at least about 5:1. In certain embodiments, the mole ratio of formaldehyde scavenging agent to reactive prepolymer is in the range of from about 0.01:1 to about 0.5:1, from about 0.5:1 to about 5:1, or from about 1:1 to about 4:1. In still other embodiments, the mole ratio of formaldehyde scavenging agent to reactive prepolymer is at least 0.05:1. In still other embodiments, the mole ratio of formaldehyde scavenging agent to reactive prepolymer is at least 5:1.

[0139] In certain embodiments, the formaldehyde scavenging agent is present in an amount from about 1% to about 50% (w/w), from about 1% to about 30% (w/w), from about 1% to about 20% (w/w), from about 5% to about 50% (w/w), from about 5% to about 30% (w/w), from about 5% to about 20% (w/w), from about 10% to about 50% (w/w), from about 10% to about 30% (w/w) or

from about 10% to about 20% (w/w) of the adhesive composition. In certain other embodiments, the formaldehyde scavenging agent is present in an amount from about 1% to about 50% (w/w) of the adhesive composition. In still other embodiments, the formaldehyde scavenging agent is present in an amount from about 2% to about 30% (w/w) of the adhesive composition.

Fire Retardants

[0140] Exemplary fire retardants include, for example, (i) phosphoric acid or a salt thereof, such as a mono-ammonium phosphate, di-ammonium phosphate, ammonium poly-phosphate, melamine phosphate, guanidine phosphate, urea phosphate, alkali metal phosphate, and alkaline earth metal phosphate, (ii) a halogenated phosphate compound, (iii) a phosphate ester, such as tri-o-cresyl phosphate and tris(2,3-dibromopropyl) phosphate, (iv) a chlorinated organic compound, such as a chlorinated hydrocarbon or chlorinated paraffin, (iv) a brominated organic compound, such as a brominated hydrocarbon, bromo-bisphenol A, tetrabromobisphenol A (TBBPA), decabromobiphenyl ether, octabromobiphenyl ether, tetrabromobiphenyl ether, hexabromocyclododecane, bis(tetrabromophthalimido) ethane, tribromophenol, and bis(tribromophenoxy) ethane, (v) a brominated oligomer or brominated polymer, such as TBBPA polycarbonate oligomer, brominated polystyrene, and TBBPA epoxy oligomer, (vi) a borate compound, such as an alkali metal borate, ammonium borate, or mixture comprising one or more of borax, boric acid, boric oxide, and disodium octaborate, (vii) aluminium materials, such as aluminium trihydrate and aluminium hydroxide, (viii) an alkaline earth metal hydroxide, such as magnesium hydroxide, (ix) an alkali metal bicarbonate, such as sodium bicarbonate, (x) an alkaline earth metal carbonate, such as calcium carbonate, (xi) antimony trioxide, (xii) hydrated silica, (xiii) hydrated alumina, (xiv) dicyandiamide, (xv) ammonium sulfate, and (xvi) a mixture of guanylurea phosphate and boric acid, such as those described in International Patent Application Publication No. WO 02/070215, (xvii) graphite, (xviii) melamine, and (xix) a phosphonate compound, such as diethyl-N,N-bis(2-hydroxyethyl) aminoethyl phosphonate; dimethyl-N,N-bis(2-hydroxyethyl) aminomethyl phosphonate; dipropyl-N,N-bis(3-hydroxypropyl) aminoethyl phosphonate; and dimethyl-N,N-bis(4-hydroxybutyl) aminomethyl phosphonate, such as described in U.S. Pat. No. 6,713,168.

[0141] In certain embodiments, the fire retardant is (i) phosphoric acid or a salt thereof, such as a mono-ammonium phosphate, di-ammonium phosphate, ammonium poly-phosphate, melamine phosphate, guanidine phosphate, urea phosphate, alkali metal phosphate, and alkaline earth metal phosphate, (ii) a phosphate ester, such as tri-o-cresyl phosphate and tris(2,3-dibromopropyl) phosphate, aluminium trihydrate and aluminium hydroxide, (iii) an alkaline earth metal hydroxide, such as magnesium hydroxide, (iv) an alkali metal bicarbonate, such as sodium bicarbonate, (v) antimony trioxide, or (vi) hydrated alumina.

[0142] In certain other embodiments, the fire retardant is Colemanite (CaB₂O₆·4(OH)·3H₂O).

Wood Preservatives

[0143] Exemplary wood preservatives include, for example, (i) chromated copper arsenate (CCA), (ii) alkaline copper quaternary, (iii) copper azole, (iv) a borate preservative compound, (v) a sodium silicate-based preservative compound, (vi) a potassium silicate-based preservative compound, (vii) a bifenthrin preservative compound, (viii) a coal-tar creosote, (ix) linseed oil, (x) tung oil, and (xi) an insecticide, such as an organochloride compound, organophosphate compound, carbamate compound, pyrethroid, neonicotinoid, and ryanoid.

[0144] The invention further relates to a process for preparing an article with the adhesive composition prepared according to the invention, further comprising a step of contacting the adhesive composition with a lignocellulosic material to provide lignocellulosic material impregnated with the adhesive composition.

[0145] By “lignocellulosic material”, it is meant a lignocellulosic material different from the heat treated dry plant meal, in particular straw, paper, cardboard, wood strands, wood veneer, wood

particles and/or wood fibers, preferably, wood strands, wood veneer, wood particles and/or wood fibers, more preferably, wood strands, wood particles and/or wood fibers. The wood particles can be chips, flakes, sawdust, powder or any other waste based on wood originating from a sawmill.

[0146] Preferably, the wood is pine wood, spruce wood, birch wood and/or beech wood, more preferably, the wood is spruce wood.

[0147] Advantageously, before, simultaneously or after the step of contacting the adhesive composition with a lignocellulosic material, the process according to invention further comprises a step of contacting the wood particles, fibers and/or strands with wax, preferably, before the step of contacting the adhesive composition with a lignocellulosic material.

[0148] When the adhesive composition and the wax are contacted simultaneously with the lignocellulosic material, the adhesive composition preferably comprises the wax.

[0149] Preferably, the wax is paraffin, carnauba wax and/or beeswax, more preferably, paraffin.

[0150] Preferably, the weight ratio of dry wax/dry lignocellulosic material is comprised between 0.04 and 5%, more preferably, between 0.1 and 2%, even more preferably, between 0.2 and 0.7%, such as, for example, between 0.3 and 0.5%.

[0151] By “contacting” the adhesive composition with a lignocellulosic material or the wood particles, fibers and/or strands with wax, it is meant preferably applying by brush or roller, mixing or spraying, more preferably, mixing or spraying, even more preferably, spraying. Indeed, by heat treating the dry plant meal, the viscosity of the adhesive composition is reduced allowing the spraying of the composition.

[0152] Advantageously, the process according to the invention further comprises a step of providing an article by curing of the adhesive composition. Preferably, the curing is implemented by heating at a curing temperature comprised between 120° C. and 250° C., more preferably, between 150° C. and 230° C., even more preferably, between 180° C. and 210° C. Preferably, the article is provided after placing the lignocellulosic material impregnated with the adhesive composition into a mold, followed by pressing and heating at the curing temperature described before.

[0153] Preferably, the article is a wood board, more preferably, the article is plywood, oriented strand board (OSB), particleboard or fiberboard, even more preferably, the article is oriented strand board (OSB), particleboard or fiberboard.

[0154] In a first embodiment, the invention relates to a heat treated dry plant meal obtained by subjecting a dry plant meal to a heat treatment at a temperature and during a treatment time sufficient to reduce the viscosity of a dispersion of said heat treated dry plant meal in water compared to the same dispersion of the non heat treated dry plant meal with water.

[0155] In a second embodiment, the invention is also directed to a heat treated dry plant meal capable of reducing the viscosity of a dispersion in water of said heat treated dry plant meal compared to the corresponding dispersion in water of the non heat treated dry plant meal.

[0156] In a third embodiment, the invention relates to a heat treated dry rapeseed or canola meal obtained by subjecting a dry rapeseed or canola meal to a heat treatment at a temperature of at least 115° C.

[0157] Advantageously, the heat treatment is performed at least 15 min, preferably, at least 30 min.

[0158] Advantageously, the heat treatment is performed during about 5 min at about 200° C., during about 15 min at about 150° C. or during about 15 min at about 120° C., preferably, during about 15 min at about 150° C.

[0159] In a fourth embodiment, the invention relates to a heat treated dry plant meal obtained by subjecting a dry plant meal to a heat treatment at a temperature of at least 150° C.

[0160] Preferably, this heat treated dry plant meal is heat treated dry sunflower, rapeseed or canola meal, more preferably, sunflower meal.

[0161] Advantageously, the heat treatment is performed at least 15 min, preferably, at least 30 min, more preferably, at least 45 min.

[0162] Advantageously, the heat treatment is performed during about 5 min at about 200° C., during about 15 min at about 175° C. or during about 30 min at about 150° C., preferably, during about 30 min at about 150° C.

[0163] The heat treated dry plant meal of the first, second, third and fourth embodiment is advantageously as described above, in particular with respect to the type of plant meal and its protein content.

[0164] The invention further relates to a heat treated dry plant meal dispersion comprising a heat treated dry plant meal and water, and having a viscosity at 20° C. of less than 1000 mPa.Math.s.

[0165] The heat treated dry plant meal is as described above. In particular, the heat treated dry plant meal is obtained by heating the dry plant meal as described above, in particular with respect to the type of plant meal and its protein content.

[0166] The heat treated dry plant meal dispersion may comprise one or more additives as described above.

[0167] Preferably, the heat treated dry plant meal dispersion consists of a heat treated dry plant meal, water and optionally one or more additive(s) as described above.

[0168] Advantageously, the heat treated dry plant meal dispersion has a solid content as described above.

[0169] Preferably, the heat treated dry plant meal dispersion has a viscosity at 20° C. of less than 500 mPa.Math.s, more preferably, of less than 400 mPa.Math.s, still more preferably, of less than 300 mPa.Math.s, even more preferably, of less than 200 mPa.Math.s.

[0170] According to a preferred embodiment, the heat treated dry plant meal dispersion has the above described viscosity and a solid content comprised between 25 wt % and 35 wt %, preferably, comprised between 27 wt % and 35 wt %, more preferably, comprised between 27 wt % and 30 wt %.

[0171] The invention also provides an adhesive composition comprising a heat treated dry plant meal, water and a reactive prepolymer, the reactive prepolymer not being soluble in water.

[0172] The adhesive composition and its ingredients (heat treated dry plant meal, reactive prepolymer, optional wax, etc.) are as described above, in particular with respect to the heat treated dry plant meal, the ingredient contents and viscosity.

[0173] More particularly, the heat treated dry plant meal is obtained from the dry plant meal as described above, in particular with respect to the type of plant meal and its protein content. Also, the reactive prepolymer is preferably an isocyanate-based prepolymer.

[0174] Advantageously, the adhesive composition according to the invention comprises heat treated dry plant meal in an amount sufficient to form an emulsion that exhibits no phase separation by visual inspection for at least 5 minutes after mixing the reactive prepolymer with the heat treated dry plant meal. In particular, presence of droplets detected by visual inspection (for example, with a microscope) means that the adhesive composition is under the form of an emulsion.

[0175] The adhesive composition may be provided under a two-part form, wherein one part comprises the reactive prepolymer and the other part comprises the heat treated dry plant meal with water. Preferably, the heat treated dry plant meal with water is a dispersion in water of the heat treated dry plant meal, as described above.

[0176] Preferably, upon curing, the adhesive composition forms a solid binder.

[0177] Advantageously, the adhesive composition according to the invention is sprayable.

[0178] The invention also relates to the use of a heat treated dry plant meal for improving processability of an adhesive composition.

[0179] The heat treated dry plant meal is as described above. In particular, the heat treated dry plant meal is advantageously a heat treated dry plant meal according to any one of the first to the fourth embodiments.

[0180] Preferably, the processability is improved by decreasing viscosity of the adhesive composition.

[0181] By “decreasing the viscosity of the adhesive composition”, it is meant that the viscosity of the adhesive composition comprising the heat treated dry plant meal is inferior to the viscosity of the same adhesive composition (i.e. at the same dry matter content) comprising the same dry plant meal not heat treated. By “inferior to”, it is meant a decrease of viscosity of preferably at least 20%.

[0182] More preferably, the processability is improved by decreasing viscosity at 20° C. of the adhesive composition to less than 1000 mPa.Math.s, preferably, less than 500 mPa.Math.s, more preferably, less than 400 mPa.Math.s, still more preferably, less than 300 mPa.Math.s, even more preferably, less than 200 mPa.Math.s.

[0183] Furthermore, the invention relates to the use of a heat treated dry plant meal for stabilizing an adhesive composition comprising water and a reactive prepolymer, the reactive prepolymer not being soluble in water.

[0184] The heat treated dry plant meal is as described above. In particular, the heat treated dry plant meal is advantageously a heat treated dry plant meal according to any one of the first to the fourth embodiments.

[0185] By “stabilizing” an adhesive composition, it is meant in particular that no change of appearance by visual inspection can be observed and/or no unpleasant odor appears at least 4 days after preparing said adhesive composition, preferably at least 5 days, such as for example, after 6 days.

Description

[0186] The invention will be better understood in the light of the following examples, given by way of illustration, with reference to:

[0187] FIG. 1 that is a scale for evaluating the visual aspect of a dispersion of a plant meal and water;

[0188] FIG. 2 that is a diagram illustrating the viscosity of dispersions prepared from water and ground rapeseed meal subjected to different thermal treatments, depending on the solid content of the dispersions expressed in weight percentage (% dispersion);

[0189] FIG. 3 that is a diagram illustrating the viscosity of emulsions prepared from pMDI and dispersions of ground rapeseed meal (heat treated or not) in water;

[0190] FIG. 4 that shows pictures of emulsions prepared from pMDI and dispersions of ground rapeseed meal (heat treated or not) in water;

[0191] FIG. 5 that is a diagram illustrating the viscosity of dispersions at a solid content of 23 wt % prepared from water and ground rapeseed meal of two different origins (GCXX-12/18 or LMXX-26/18) subjected or not to a thermal treatment, depending on the granulometry d(50);

[0192] FIG. 6 that is a diagram illustrating the viscosity of dispersions prepared at a laboratory scale from water and ground rapeseed meal of two different origins (GCXX-12/18 or LMXX-26/18) subjected or not to a thermal treatment and having a granulometry d(50) of 30 µm, depending on the solid content of the dispersions;

[0193] FIG. 7 that is a diagram illustrating the viscosity of emulsions prepared from pMDI and dispersions at a solid content of 23 wt % in water of ground rapeseed meal of two different origins (GCXX-12/18 or LMXX-26/18) subjected or not to a thermal treatment, depending on the granulometry d(50);

[0194] FIG. 8 that shows pictures of emulsions prepared from pMDI and dispersions in water of ground rapeseed meal of two different origins (GCXX-12/18-01 or LMXX-26/18-02) subjected or not to a thermal treatment, depending on the granulometry d(50);

[0195] FIG. 9 that is a diagram illustrating the internal bonding (IB) and density of particle boards made from pMDI and dispersions in water of ground rapeseed meal (different origin: GC or LM,

different granulometry: 30 μm , 100 μm or 250 μm , and heat treated or not), IB being represented by the bars and density by the line;

[0196] FIG. **10** that is a diagram illustrating the thickness swelling (TS) after 24 h and density of particle boards made from pMDI and dispersions in water of ground rapeseed meal (different origin: GC or LM, different granulometry: 30 μm , 100 μm or 250 μm , and heat treated or not), TS being represented by bars and density by the line;

[0197] FIG. **11** that shows pictures of dispersions prepared from water and non-treated rapeseed meal after 4 days (left picture), and from water and heat-treated rapeseed meal after 7 days (right picture);

[0198] FIG. **12** that is a diagram illustrating the concentration of microorganisms per mL of a dispersion at a solid content of 20 wt % prepared from water and heat treated rapeseed meal, depending on the days after dispersing;

[0199] FIG. **13** that is a diagram illustrating the viscosity of dispersions prepared from water and ground rapeseed meal (heat treated or not), depending on the solid content of the dispersions;

[0200] FIG. **14** that is a diagram illustrating the viscosity of emulsions prepared from pMDI and dispersions in water of heat treated then ground rapeseed meal (weight ratio pMDI/meal being 2 or 4), depending on the concentration of the meal in the dispersions;

[0201] FIG. **15** that shows pictures of emulsions prepared from pMDI and dispersions in water of heat treated then ground rapeseed meal, depending on the concentration of the meal in the dispersions (20, 23, 25, 27 or 30 wt %), and the pMDI/meal weight ratio (2 or 4);

[0202] FIG. **16** that is a diagram illustrating the internal bonding (IB) and density of particle boards made from wax and pMDI with dispersions in water at a solid content of 20 or 27 wt % of ground rapeseed meal (optionally heat treated before grinding), using two different formulations (bars with diagonal lines or with filling points), IB being represented by the bars and density by the line;

[0203] FIG. **17** that is a diagram illustrating the thickness swelling (TS) and density of particle boards made from wax and pMDI with dispersions in water at a solid content of 20 or 27 wt % of ground rapeseed meal (optionally heat treated before grinding), using two different formulations (bars with diagonal lines or with filling points), TS being represented by the bars and density by the line;

[0204] FIG. **18** that is a diagram illustrating the viscosity of dispersions prepared at a pilot scale from water and ground rapeseed meal (optionally heat treated before grinding), depending on the solid content of the dispersions;

[0205] FIG. **19** that is a diagram illustrating the viscosity of emulsions prepared from pMDI and dispersions in water of ground rapeseed meal (optionally heat treated after grinding), depending on the concentration of the meal in the dispersions;

[0206] FIG. **20** that shows pictures of emulsions prepared from pMDI and dispersions in water of ground rapeseed meal (optionally heat treated after grinding), depending on the concentration of the meal in the dispersions, and the pMDI/meal weight ratio;

[0207] FIG. **21** that is a diagram illustrating the internal bonding (IB) and density of particle boards made from wax and pMDI with dispersions in water at a solid content of 20 or 27 wt % of ground rapeseed meal (optionally heat treated after grinding), using two different formulations (bars with diagonal lines or with filling points), IB being represented by the bars and density by the line;

[0208] FIG. **22** that is a diagram illustrating the thickness swelling (TS) and density of particle boards made from wax and pMDI with dispersions in water at a solid content of 20 or 27 wt % of ground rapeseed meal (optionally heat treated after grinding), using two different formulations (bars with diagonal lines or with filling points), TS being represented by the bars and density by the line;

[0209] FIG. **23** that is a diagram illustrating the internal bonding (IB) of OSB made from wax, pMDI and dispersions in water at a solid content of 20 wt % of ground rapeseed meal heat treated ("rapeseed TT", granulometry d(50) of 80 μm) or not (granulometry d(50) of 95 μm) in the face,

and made from wax and pMDI alone or with dispersions in water at a solid content of 20 wt % of ground rapeseed meal heat treated (“rapeseed TT”, granulometry d(50) of 80 µm) or not (granulometry d(50) of 95 µm) in the core;

[0210] FIG. **24** that is a diagram illustrating the thickness swelling (TS) of OSB made from wax, pMDI and dispersions in water at a solid content of 20 wt % of ground rapeseed meal heat treated (“rapeseed TT”, granulometry d(50) of 80 µm) or not (granulometry d(50) of 95 µm) in the face, and made from wax and pMDI alone or with dispersions in water at a solid content of 20 wt % of ground rapeseed meal heat treated (“rapeseed TT”, granulometry d(50) of 80 µm) or not (granulometry d(50) of 95 µm) in the core;

[0211] FIG. **25** that is a diagram illustrating the viscosity of dispersions at a solid content of 20 wt % prepared from water and ground sunflower meal (heat treated or not), depending on the pH of the dispersions;

[0212] FIG. **26** that is a diagram illustrating the viscosity of dispersions at a solid content of 30 wt % prepared from water and ground rapeseed meal (heat treated or not), depending on the pH of the dispersions;

[0213] FIG. **27** that is a diagram illustrating the relative strength development of adhesive compositions (Formulas 5A, 5B, 7A and 7B) depending on the press time;

[0214] FIG. **28** that is a diagram illustrating the relative strength development of adhesive compositions (Formulas 1, 2, 5 and 6) depending on the press time;

[0215] FIG. **29** that is a scheme of a sterilization reactor.

EXAMPLE 1: MATERIALS AND METHODS

Materials

[0216] All raw material used are listed in Table 1.

TABLE-US-00001 TABLE 1 description of the raw material used in the Examples, the percentages being weight percentages

Protein	Oil	Name	content	content	Supplier	Manufacturing process
Wheat	draff	27%	6.4%	Aso	By-product from the manufacture of Nutrition alcohol by distillation from wheat	Corn draff
21%	13.9%	Aso	By-product from the manufacture of Nutrition alcohol by distillation from corn	Linseed meal	34%	n.d.
Saipol	From crushing process of flaxseed	Soy meal	51%	0.7%	ADM	From crushing process of soy seeds extracted with hexane
Sunflower meal	37%	1.3%	Saipol	From crushing process of sunflower and meal	Bassens	seeds with hexane pellets
Rapeseed meal	35%	2.35%	Saipol	From crushing process of rapeseed pellets seeds with hexane	Rapeseed cold-	18%
11.7%	Saipol	From crushing process of rapeseed press cake seeds with no hexane (expeller)	Soy protein	70%	n.d.	Efos concentrate
Soy protein	90%	n.d.	Efos isolate	ISP920H	Brassica	40%
2.25%	Saipol	From crushing process of <i>Brassica carinata</i> meal	<i>Carinata</i>	seeds with hexane		

[0217] Unless otherwise specified, the meal is in loose (powder) form.

Product Characterization

Dry Content

[0218] The material is placed in an aluminum cup with an amount of about 1.5 g. The exact weight is measured with an analytical balance with 4 digits after the decimal point. Then the cup is placed in a fan ventilated oven at 105° C. for 3 hours. Fan & valve aperture are set at 100% to accelerate water evaporation. Dry content is obtained by measuring the weight of the cup after being removed from the oven immediately or after conditioning in a desiccator to get sample at room temperature. Calculation of dry content is obtained by the calculation of material loss thanks to (1) and (2):

$$[00001] \quad m_{\text{loss}} = m_0 - m_1 \quad (1)$$

[0219] Where m.sub.0 and m.sub.1 are respectively the weight of the cup before being placed in the oven and after being placed in the oven.

$$[00002] \quad D C \% = 1 - \frac{m_{\text{loss}}}{m_{\text{sample}}} \quad (2)$$

[0220] Where DC %, m.sub.loss and m.sub.sample are respectively the dry content, the weight of

material loss and the weight of sample tested (m.sub.0–m.sub.cup).

[0221] An alternative method consists in using a moisture analyzer (Sartorius M50) to get the dry content or the moisture content. It enables to have the “atro” mode (i.e. moisture content in weight parts over dry composition) required for wood-based panel manufacturing. This method is also faster (i.e. few minutes versus 3 hours) but allows to test only one sample per round. The moisture analyser is preferably used for wood DC % determination prior to wood panel making.

Particle Size Determination

[0222] The particle size of ground samples is measured using a Malvern laser granulometer Mastersizer 3000. Sample material is injected through the dry injection tool of the analyzer. Refractive index and model used are respectively 1.52 and Mie to determine the particle size density profile. The main value of interest is the d50 previously described.

Viscosity Measurement

[0223] The viscosity of the dispersions and adhesive compositions was measured by using a rheometer (Haake MARS 40 by Thermo Scientific) equipped with a Pelletier heating/cooling system. The measurement device is a plate/plate system (20 mm diameter, 1 mm gap). Measurement is done with a dynamic mode (shear rate of 10 s.sup.–1, 50% shear and 20° C. temperature). The viscosity value was automatically determined after 5 minutes of measurement using Rheowin Job manager version 4.81.0000 (Haake Mess Tech, Germany).

Emulsion Quality Assessment

[0224] Emulsion quality is determined directly after its fabrication using a microscope Leica DM750M with a camera Leica MC170 HD. The microscope lens used for the magnification of the sample is $\times 10$ in transmittance mode. The assessment is done by direct comparison of picture with the same magnification in a qualitative manner.

Panels Characterization

[0225] Laboratory boards (200×200 mm) are cut to get samples for water resistance and internal bonding (IB).

[0226] For pilot board: pilot boards are cut to get samples for water resistance and IB. Water resistance and IB are done respectively following standard ISO EN 317:1993 and ISO EN 319:1993 to get Thickness Swelling (TS) and IB. Apparatus used for all these measurements is the Imal (Italia), IBX700 model. The test results are mean values with their standard deviation. Each sample is measured and weighed to get its density thanks to the IBX700.

Raw Material Microbiological Stability

[0227] In order to quantify microbiological stability, three tests were implemented to follow the evolution of the odor, surface appearance and biological activity of dispersions prepared from plant meals. At laboratory scale, approximately 300 g of dispersion were prepared in a polypropylene beaker of 400 mL and closed by a parafilm to avoid water evaporation. The three tests were monitored over time.

a) Odor

[0228] The perception of odor is subjective, and the following scale was used to rank odor level:

[0229] 1: Standard odor of a fresh dispersion; [0230] 2: Slightly modified odor but not unpleasant;

[0231] 3: Strongly modified odor but not unpleasant; [0232] 4: Slightly modified unpleasant odor;

[0233] 5: Strongly modified unpleasant odor.

[0234] For the “Bad” sample, significant development of microorganisms (molds+bacteria) on the surface can be seen.

b) Visual Appearance

[0235] A scale for the appearance was implemented in order to have a qualitative evaluation of the visual evolution and is provided on FIG. 1.

[0236] For the “OK” sample, the dispersion of plant material is identical to T0 (fresh dispersion), in static state, with a liquid top layer having a light brown color.

[0237] For the “Fair” sample, the dispersion begins to contain solid aggregated (molds) on the top.

c) ATP-Metry

[0238] The last test to quantify the microbiological activity of the dispersion over time is ATP-metry.

[0239] ATP (adenosine triphosphate) is a molecule synthesized in all living cells (bacteria, yeasts, mammalian cells . . .). When ATP breaks down in cells, it releases energy used by “cellular machines”, allowing them to work. It is their reserve of energy.

[0240] ATP can be quantified by a bioluminescence reaction, i.e. by light production caused by the chemical reaction between ATP and an enzymatic complex, luciferin-luciferase. This enzymatic complex, in presence of oxygen and magnesium, transforms the energy released by the hydrolysis of ATP into light. This light is measured by a luminometer, and the amount of light produced is proportional to the amount of ATP present in the measured sample, and thus living cells. Method, material and reagents used to realize this test were provided by GLBiocontrol (France).

[0241] The method follows these steps: approximately 30 g of dispersion were placed in a tube and centrifuged (model Sorvall ST8 from Thermo Scientific) at 3000 rpm for 10 minutes. This step was repeated twice with the supernatant for solid/liquid separation. A volume V of the final supernatant (no more than 2 mL) was placed in a sterile syringe including a sterile 0.45 µm porosity filter screwed at the output. The plunger of the syringe was then pushed to allow the sample to pass through the filter. Then 4 droplets of the reagent Dendridiag® IW (from GLBiocontrol, France) were sucked up by the syringe through the filter and then the contents of the syringe were pushed into a measuring tube provided for a luminometer. This tube was then tested into a luminometer Kikkoman PD30 to obtain the value R1 (in RLU: Relative Light Unit). Then, one droplet of the reagent named Standard 1000 (from GLBiocontrol) was added into the tube, and a new measurement with the luminometer is carried out to obtain the value R2 (in RLU). Finally, by using the equations described below, the bacteria equivalent concentration per milliliter was obtained.

[00003] $\text{Standard} = \frac{R2 - R1}{1000} [\text{ATP}] = \frac{R1}{\text{Standard} \times V}$ [0242] With: [0243] Standard=Value of the standard (RLU/pg ATP) [0244] R1=result on sample (RLU) [0245] R2=result on sample+standard (standard addition method) (RLU) [0246] V=filtered volume in milliliter [0247] [ATP]=ATP concentration (pg ATP/mL) [0248] 1 picogramme (pg) of ATP is about 1000 bacteria.

EXAMPLE 2: PREPARATION OF ADHESIVE COMPOSITIONS FROM HEAT GROUND DRY MATERIAL AT LABORATORY AND PILOT SCALES

Preparation of Heat Treated Ground Dry Material at Laboratory Scale

[0249] At the laboratory scale, raw material was ground using a laboratory grinder from Retsch model ZM 200. This equipment allows to get well define granulometry profile characterized by the d50 value (maximum size of 50% of the smallest particles in weight).

[0250] A ring sieve (80, 120 and 200 µm openings) is used to tailored particles size distribution and especially the d50 value in combination with mill speed (12000 and 18000 rpm). Higher is the sieve opening higher is the d50 and the lower is the speed, the higher is the d50.

[0251] Thermal treatment was applied on raw materials in an oven (UF75, Memmert, Germany) by controlling temperature and treatment time. To ensure the homogeneity of the treatment in the oven, raw materials were spread in thin layers (9±1 mm) on plates. Once the treatment is done, plates containing treated material were removed from the oven to let them cool down and equilibrate under room condition, and then was kept in a closed container for further analysis and use.

[0252] After the thermal treatment, the moisture content needs to be close to 0% (1% to 0%). If the value is higher, the thermal treatment was not done correctly and needs to be rerun. All products tested in the present Examples always had a moisture content lower than 1 wt %.

Preparation of Heat Treated Ground Dry Material at Pilot Scale

[0253] Two different processes were considered.

Grinding and Thermal Treatment:

[0254] Materials, in pellet shape, were first ground at the desired particle size (d50). Pilot plant is

equipped with an impact mill grinder including a classifier from Hosokawa, model 200ZPS. [0255] After the classifier, the product is moved to Guedu dryer for thermal treatment. Guedu dryer is from De Dietrich Process System with a tank of 1600 L using a heating system with a double jacket and a powder impeller. The impeller ensures good product recirculation in the dryer and avoid particles agglomeration. The thermal treatment is realized by a temperature ramp and mixing speed.

Thermal Treat and Grinding:

[0256] In this variant, pellets are transferred to the Guedu dryer first for thermal treatment with the same protocol (heating ramp). This step allows to treat material before grinding and contributes to a first particle size reduction (pellet deagglomeration). Then the heat treated pellets are transferred to the mill (model 200ZPS, Hosokawa) for grinding operations to get the desired d50.

[0257] After the thermal treatment, the moisture content needs to be close to 0% (1% to 0%). If the value is higher, the thermal treatment was not done correctly and needs to be rerun. All products tested in the present Examples always had a moisture content lower than 1 wt %.

Preparation of Adhesive Composition (Emulsion)

Preparation of a Heat Treated Dry Plant Meal Dispersion

[0258] To prepare the adhesive composition, a dispersion was first prepared by adding from 20 wt % to 30 wt % of heat treated ground material (based on the targeted solid content accounting of the residual water adsorbed in the material and determined previously with the dry content test) in tap water at ambient temperature (20° C.) in a polypropylene beaker of 100 mL. The dispersion is mixed using a deflocculating blade (10 blades of 49 mm diameter, rod of 7 mm diameter) for two minutes.

Preparation of an Emulsion

[0259] The emulsion is prepared by adding polymethylene polyphenyl diisocyanate (pMDI) (Suprasec® 2085, Huntsman). An adequate amount of pMDI is added to achieve the targeted weight ratio pMDI/meal (dry/dry). This ratio is chosen to achieve a certain emulsion quality & viscosity. The mixing is pursued as described previously with a higher mixer speed (1000 rpm) for two minutes. When the emulsion looks homogenous (no phase separation for 10 minutes by visual inspection between pMDI and meal dispersion) the adhesive composition is ready for next step.

Preparation of Particle Boards at Laboratory Scale

[0260] To manufacture laboratory particle boards, the adhesive composition is blended with wood particles using a planetary blender (pluton, mineapolis) with a paste impeller. An amount of 700 g of wood is introduced in the blender tank after moisture content determination which is expected to be around 4.5% atro. Wax emulsion (Wax in aqueous emulsion Michem Wood Coating 70LV from Michelman with a solid (wax) content of 50 wt % and a water content of 50 wt %) is added first to the wood. The adequate amount of wax emulsion is weighed and introduced with a syringe while the blender is running with the lowest speed. The standard value is 0.5 parts of dry wax for 100 parts of dry wood, in weight. After the addition of wax, the adhesive composition is added to the wood furnish in the same manner than the wax (syringe while the blender is running). An extra 5 min mixing time is applied to ensure homogenous distribution of the glue with the wood furnish.

[0261] When a target moisture for wood furnish before pressing is required, water could be added to the wood before the wax. The adequate value is calculated by considering the targeted value to reach and removing from this calculation the water already adsorbed in the wood, water contained in the wax emulsion and the water contained in the adhesive composition. Standard targeted Mat (resin impregnated wood before hot pressing) moisture content value is about 10% atro. The real value will be checked after blending with the moisture analyzer equipment (faster than the oven dry test method). At the end of this step, an impregnated wood sample ready to form a laboratory size particle board is obtained.

[0262] A 200×200×200 mm wood forming box is filled with impregnated wood (Wood+Adhesive+water+wax) sample weighed to achieve a particle board with a density of 650

kg/m³ at a thickness of 12 mm (final thickness is obtained thanks to a stainless-steel mold of 12 mm thick). After that, impregnated wood is cold pressed by hand to form a mat. The mat is hot pressed to a thickness of 12 mm for 5 minutes at a press platen temperature of 180° C. After pressing, thickness and weight of the particle board samples were measured and recorded. Then, the particle board was placed in a conditioning room at 20° C. and 65% r.h. (relative humidity) for at least one day. For each adhesive composition, 2 boards were made.

Preparation of OSB (i.e. Oriented Strand Boards) at Pilot Scale

[0263] At pilot scale, strands (large particles for OSB) and not only particles can be used. In addition to that, tri-layer boards (instead of monolayer) closer to industrial standards can be produced accurately. The strands (spruce) used came from Fraunhofer WKI (Braunschweig, Germany) with a moisture content of 4% (ATRO). An amount of 7 to 9 Kg of wood strands was placed into a rotary blender (Coil 6×3 m laboratory sized running at 18 rpm) depending of batch size required (core layer or surface layer). The wax and the water are injected thanks to a spray nozzle within the blender. The adhesive composition is injected thanks to a spinning disc (from Coil, USA) turning at 9600 rpm. The adhesive composition injection time is about 2 minutes depending on the viscosity of the product thanks to a peristaltic pump as the wax emulsion. The adhesive composition (emulsion of pMDI with the dispersion of ground plant meal) is obtained inside the nozzle by using the dual port atomizer from Coil which have 2 ports for liquids injection and a blending area inside the spinning disc before liquid atomization.

[0264] Tri-layer boards are produced thanks to 2 batches, one for face layer (top and bottom) and one for core layer (middle). The differences of these 2 batches are the amount of adhesive composition with respect to dry wood, and the moisture content which needs to be low in the core layer (4-6% ATRO) and high in the face layer (12-15% ATRO).

[0265] Adequate quantity of impregnated strands is weighed to achieve a density of approximately 580 kg/m³ at a thickness of 15 mm with a surface/core ratio of 60/40. Forming box size is 600×600×200 mm to make the mat. After wood/adhesive sample addition, the mat was cold pressed by hand. The mat is then placed into a heating press from Imal (Italia) (platen of 1000×1000 mm) and pressed to a thickness of 15 mm and cooked at 210° C. at 8 s/mm (120 s). After pressing, thickness and weight of the OSB is measured and recorded. Then the board is placed in a conditioning room at 20° C. and 65% r.h. for at least one day. For each adhesive composition, 2 boards are made.

EXAMPLE 3: DETERMINATION OF A LOWER TEMPERATURE RANGE

[0266] In a first step, it was evaluated if there is a minimum temperature to apply a thermal treatment and quantify any thermal barrier to overpass to get the treatment happening. The efficiency of the treatment is based on viscosity reduction versus a reference material not treated. Temperatures from room temperature to 105° C. were tested. An overnight time for the treatment was chosen to ensure enough time to get something quantifiable. Viscosity was checked at solid content from 25 to 30 wt % to quantify benefits even if the current usage is closer to 20 wt % for value discrimination purpose. Dispersions having different solid contents prepared according to Example 2 (laboratory scale, dispersions obtained before adding PMDI) from ground rapeseed meal having a d₅₀ at 30 µm were tested. The results are shown on FIG. 2.

[0267] The results show that a treatment done at 50° C. and 70° C. has an insignificant influence on viscosity of these dispersions. However, the thermal treatment at 105° C. overnight dramatically reduces the viscosity of ground meal dispersions whatever is the solid content (% dispersion). The biggest drops obtained thanks to thermal treatment were observed at high solid content. 105° C. appears to be closed to the lower temperature limit to get an efficient thermal treatment.

[0268] To check if the benefit of viscosity reduction was kept while an emulsion is done, a dispersion of rapeseed meal at 25 wt % in water and pMDI were blended with a weight ratio of 2.67 (pMDI/dry rapeseed meal), as described in Example 2 ("Preparation of adhesive composition"). The viscosity of emulsions prepared from non-treated rapeseed meal and from heat

treated rapeseed meal (105° C., overnight) was measured, the results being shown on FIG. 3.
 [0269] The results highlight the benefit of thermal treatment to reduce the viscosity of the emulsion and thus the processability of the product for its application on wood.
 [0270] In order to characterize the emulsion quality, same samples were placed under microscope (magnification $\times 10$) and pictures were taken with the digital camera. The results are shown on FIG. 4.

[0271] It can be observed that very small pMDI droplets (white circles) are well distributed, darker particles correspond to rapeseed product (core and hulls). Once a thermal treatment is applied, one can observe a slight coloration of the rapeseed particles and an increase of the size of pMDI droplets. It means that the thermal treatment slightly decreases the emulsification power of the meal but keeping it at a good level meaning that a stable emulsion of pMDI is formed. Thermal treatment does not affect significantly emulsification power. Minor changes observed could be linked to the impact of viscosity of the dispersion that could affect easiness to get an emulsion.

EXAMPLE 4: THERMAL TREATMENT OPTIMIZATION

[0272] To optimize the thermal treatment, a small quantity of the same ground rapeseed meal was conditioned in an oven only 2 h versus overnight, and dispersions were prepared with a solid content of 25 wt % (process according to Example 2, laboratory scale). The viscosity of the dispersions was measured and is indicated in Table 2.

TABLE-US-00002 TABLE 2 viscosity of dispersions prepared from ground rapeseed meal

Viscosity	Temperature	Time (25% dispersion)	No treatment
1252 mPa .Math. s	105° C.	2 h	1252 mPa .Math. s
200 mPa .Math. s	105° C.	12 h	

[0273] At 105° C., 2 h was not enough to get a positive impact especially versus a 12 h time as in Example 3. It highlights the need to move to higher temperature to shorten treatment time. The tests at higher temperatures were handle on rapeseed meal and pellets: heat treatment was done on the ground meal (after grinding) and on the meal pellets (before grinding). The viscosity of the dispersions (25 wt % solid content) prepared from rapeseed meal and rapeseed meal pellets are indicated in Tables 3 and 4 respectively.

TABLE-US-00003 TABLE 3 viscosity of dispersions prepared from rapeseed meal

Rapeseed meal	Process Viscosity, 25% Relative	Process time	dispersion viscosity, temperature (min) (mPa .Math. s)	Uncertainty %
20° C.	0 495	74.25	100.0%	120° C. 15 387 58.05 78.2%
30 322	48.3	65.1%	45 280 42 56.6%	150° C. 15 161 24.15 32.5%
30 80	12 16.2%	45 64 9.6 12.9%	200° C. 5 67 10.05 13.5%	

[0274] The first value (20° C., Omin) corresponds to the untreated sample. One can observe a different value than in Table 2 even if it is the same raw material origin obtained at a different period of time (495 mPa.Math.s vs. 1252 mPa.Math.s). This difference is not surprising as Rapeseed is a natural product, its composition and characteristics are changing based on, for example, season, origin, process parameter, and induce viscosity variations.

[0275] At 120° C. for a process time of 30 and 45 minutes, viscosity values are in between 322 to 280 mPa.Math.s, evidencing a clear trend of viscosity reduction versus time to 57% of initial viscosity value.

[0276] At 150° C., viscosity values are even lower and in between 161 and 61 mPa.Math.s. In other words, a treatment at 150° C. for 15 min is more efficient than 120° C. for 45 min. At 150° C. the viscosity drop from 160 to 80 mPa.Math.s by increasing treatment time from 15 to 30 min which correspond to 32% to 16% of the initial value. When the treatment time is increased to 45 min, viscosity decreases to 67 mPas i.e. 13% of the initial value. 150° C. and 30 min appears to be a good compromise between time and viscosity decrease at this stage.

[0277] At 200° C., only one treatment time was tested (5 min), giving a value of 67 mPa.Math.s (13.5% of the initial value) which is equal to 150° C. and 45 min treatment. It is an even more efficient condition. However, at the laboratory scale it will be hard to control a so short treatment time. This condition is more suitable for a continuous process rather than a batch process.

[0278] Same approach was conducted on Rapeseed meal pellets:

TABLE-US-00004 TABLE 4 viscosity of dispersions prepared from rapeseed meal pellets
Rapeseed meal pellets Process Viscosity, 25% Relative Process time dispersion viscosity,
temperature (min) (mPa .Math. s) Uncertainty % 20° C. 0 470 70.5 100.0% 120° C. 15 272 40.8
57.9% 30 222 33.3 47.2% 45 204 30.6 43.4% 150° C. 15 99 14.85 21.1% 30 58 8.7 12.3% 45 45
6.75 9.6%

[0279] The reference product (20° C., Omin) has a viscosity of 470 mPa.Math.s, which is close to the previous value which correspond to the experimental error using the same material.

[0280] At 120° C., viscosity values are in between 272 and 200 mPa.Math.s (58 to 43% of the initial value). It appears that viscosity reduction obtained on meal pellets is more significant than on ground meal.

[0281] At 150° C., viscosity reduction is close to 10% (for 30 and 45 min) which is close to what the values obtained with ground meal. It seems that the advantage of pellet shape versus ground meal depends on temperature.

[0282] No measurement was done at 200° C. as it becomes too high. Pellets start to burn.

[0283] In conclusion, on powder meal, preferred temperature range will be 150 to 200° C. to get a significant viscosity reduction. Higher is the temperature, lower is the time required to get the lowest viscosity possible. On meal pellets, even if heat treatment is better for intermediate temperature and intermediate performance, the preferred temperature range will also be 150 to 200° C.

EXAMPLE 5: THERMAL TREATMENT ROBUSTNESS: 150° C. DURING 30 MIN FOR VARIOUS PLANT MEALS

[0284] A clear benefit of thermal treatment has been observed with rapeseed meal with an optimum temperature in between 150 to 200° C. and less than 30 min. The same set of experiments was carried out with another plant meal which is sunflower meal, dispersions being prepared with a solid content of 25 wt % (process according to Example 2, laboratory scale). Treatments have been done only on ground sunflower meal, the viscosity of the dispersions being indicated in Table 5.

TABLE-US-00005 TABLE 5 viscosity of dispersions prepared from sunflower meal Sunflower meal Process Viscosity, 25% Relative Process time dispersion viscosity, temperature (min) (mPa .Math. s) Uncertainty % 20° C. 0 64000 17920 100.0% 150° C. 15 44500 12460 69.5% 30 14800 4144 23.1% 45 1796 502.88 2.8% 175° C. 15 211 59.08 0.3% 30 70 19.6 0.1% 200° C. 5 190 53.2 0.3% 15 46 12.88 0.1%

[0285] The reference value is much higher for sunflower meal than for rapeseed meal for the same dry content, 64000 mPa.Math.s versus 1200-500 mPa.Math.s respectively.

[0286] At 150° C., a significant viscosity reduction to 23% and 3% of the initial value was observed for 30 and 45 min treatment time respectively. There is a clear link between treatment time and viscosity reduction.

[0287] At 175° C., the effect is even more pronounced and it is possible to get viscosity values equivalent to heat treated rapeseed even if the reference viscosity of the sunflower meal dispersion is much higher. At 200° C., equivalent viscosity values could be achieved in a shorter time.

[0288] In conclusion, Sunflower meal needs a higher temperature to get viscosity values similar to rapeseed. However, with the same thermal treatment (same time and temperature), same level of viscosity reduction could be obtained. At 150° C. for 30 min, rapeseed exhibits a viscosity reduction of 12% while sunflower exhibits a viscosity reduction of 23%.

[0289] Even if thermal treatment needs to be adapted based on raw material origin, a temperature of 150° C. and time of 30 min was set for comparison purpose. At this setting any possible improvement brought by thermal treatment could be observed.

[0290] To check quickly the efficiency of thermal treatment, a quick test could be moisture measurement. In general, rapeseed meal pellet contains about 11 wt % moisture.

[0291] When pellets are ground, the moisture content decreases based on the micronization

process. This moisture content could range from 10 wt % to 6 wt % for an impact mill micronizer. This range is usually linked to the final particle size reduction: the dryer, the smaller. After the thermal treatment, the moisture content needs to be close to 0% (1% to 0%). If the value is higher, the thermal treatment was not done correctly and needs to be rerun. All products tested in the present Examples always had a moisture content lower than 1 wt %.

[0292] Dispersions with all samples (with and without thermal treatment) were made at several solid contents to account of their different behavior in term of processability while used (as observed in between Sunflower and Rapeseed). Results are summarized in the following Table 6, wherein the reduction of viscosity between untreated and heat treated material is indicated in parenthesis.

TABLE-US-00006 TABLE 6 Viscosity values (mPa .Math. s) of raw materials in water dispersion for several solid contents with viscosity decrease percentage

Solid content (wt %)	5%	10%	15%	20%	25%	30%	35%
Wheat draff No treatment	58	275	2520	150° C./30 min	48	210	800
(-17%) (-24%) (-68%)							
Corn draff No treatment	60	200	930	5100	150° C./30 min	60	48
(-70%) (-95%) (-95%)							
Flaxseed meal No treatment	190	2920	150° C./30 min	20	565	(-89%) (-81%)	
Soy meal No treatment	160	370	2200	150° C./30 min	50	40	110
(-69%) (-89%) (-95%)							
Rapeseed No treatment	410	1600	cold-press	150° C./30 min	40	900	cake
(-90%) (-44%)							
(expeller) Soy protein No treatment	33	670	270000	concentrate	150° C./30 min	14	62
(-58%) (-91%) (-99%)							
Soy protein No treatment	40	15100	isolate	150° C./30 min	17	50	(-57%) (-99%)
<i>Brassica</i> No treatment	35	260	65000	<i>carinata</i> meal	150° C./30 min	35	40
(-84%) (-99%)							

[0293] One can observe different behaviors from one product to another. All products (containing at least 15 wt % of proteins) exhibit a benefit of thermal treatment which leads to a dramatic reduction of viscosity. It appears that this reduction of viscosity is even more impressive for high protein content materials like soy protein concentrate and isolate. Aside of the impact of thermal treatment, it appears that material origin has an impact on viscosity while dispersed in water and, for this reason, solid content range has been adapted each time. High protein content material like soy protein isolate, soy protein concentrate, and soy meal can be processed only at a maximum of, respectively, 5%, 15% and 15% solid contents in weight percentages. When a thermal treatment is applied, this solid content could be increased up to respectively 10%, 20% and 20%. Other meals like rapeseed cold press cake and *Brassica carinata* meal can be processed at a higher solid content without thermal treatment up to respectively 15% and 20% and with thermal treatment up to respectively 20% and 20%.

[0294] Linseed meal exhibits a different behavior with a very low solid content to be processable (5 wt %).

[0295] In conclusion, the benefits of thermal treatment are not equal from one material to the another. It correlates partially to protein content in this investigated range.

EXAMPLE 6: THERMAL TREATMENT ROBUSTNESS: 150° C. DURING 30 MIN FOR RAPESEED MEAL

[0296] In order to confirm thermal treatment advantages, various rapeseed meal origins (lot number: GCXX-12/18 and LMXX-26/18) were tested with the same protocol as described in Example 2 (150° C. and 30 min, laboratory scale until preparation of dispersion) to check if these parameters duly always decrease viscosity or need to be adjusted on a material origin. The protocol (150° C. and 30 min) has been re-applied on various grinded meal (various origin & granulometry) at 2 solid contents (23 wt % & 25 wt %). Only the results at 23 wt % solid content are presented on FIG. 5 since same trends were found for both solid contents.

[0297] One can observe that meal origin affect viscosity (it has been confirmed with other meal origin). Granulometry d50 affect dramatically viscosity and processability consequently.

[0298] When a thermal treatment is applied, it becomes harder to differentiate, based on viscosity values, meal origins or the impact of granulometry. It means that, thermal treatment, aside of

bringing a general processability improvement thanks to viscosity reduction, could “erase” the impact of granulometry and meal origin.

[0299] To extend these statements to other conditions, dispersions were prepared at different solid contents from rapeseed meals of different origins (GCXX-12/18 and LMXX-26/18) having a d50 at 30 µm with and without thermal treatments. The viscosity of the dispersions was measured and is presented on FIG. 6. Viscosity reduction was observed with solid content from 20 wt % to 27 wt % enabling to extent solid content range of usage up to 30 wt % (it was not possible to achieve a low enough viscosity to prepare dispersions at 30 wt % solid content from non-treated rapeseed meal).

[0300] Then, dispersions were mixed with pMDI with the weight ratio pMDI/dry ground rapeseed meal=2.67 with a 23 wt % solid content dispersion of ground rapeseed meal in water, following the method previously described in Example 2 (laboratory scale). The viscosity of the resulting emulsions was measured and the results are shown on FIG. 7.

[0301] The same observation as above can be made. Thermally treated meal reduces viscosity of emulsions and improves the processability, especially the spray ability of the adhesive on wood particles whatever is the granulometry. On the opposite of dispersion results, d50 appears to slightly impact viscosity of the emulsion still in a reasonable manner enabling to increase d50 up to 100 µm.

[0302] Then, emulsion qualities were characterized by microscopy (magnification ×10) and the results are shown on FIG. 8.

[0303] As previously reported in Example 3, granulometry d50 variation can be observed directly on microscope. Aside of 30 µm d50 particle size emulsion, it becomes harder to observe pMDI droplets and emulsion quality. However, one can observe that pMDI droplets become bigger when a thermal treatment is applied. On FIG. 8, all emulsions were made with the same solid content (23 wt %).

[0304] Some particle boards were made with these emulsions as described in Example 2 (1.76 part of pMDI, 0.66 dry part of ground rapeseed meal at 23 wt % solid content in water, and 0.55 dry part of wax for 100 parts of dry wood, the parts being weight parts). The targeted moisture content of the mat is 10% atro. Internal bonding and density of the particle boards were measured and are shown on FIG. 9 where the bars correspond to IB and the line to density.

[0305] From mechanical performance, when a thermal treatment is applied (darker bars) all values become at least equal to reference without thermal treatment and variability is reduced.

[0306] Thickness swelling of the same particle boards was also measured and the results are shown on FIG. 10. Water resistance test exhibits same trends as mechanical performance. A slight improvement of performance could be observed when meal is heat treated.

[0307] Finally, all these laboratory results confirm interest of thermally treated ground rapeseed meal. The main benefit is a decrease of viscosity, which allows an improvement in the ease of use of adhesive. It also enables the usage of higher granulometry material and reduce meal origin impact on viscosity and thus the reduction of lot-to-lot variation. Last, mechanical performances and water resistance of boards made with thermally treated meal are kept or slightly improve at laboratory scale.

EXAMPLE 7: EVALUATION OF MICROBIOLOGICAL STABILITY

[0308] To check the impact of thermal treatment, odor and surface aspect evolutions were monitored at a laboratory scale. It enables to assess the impact of thermal treatment on product stability. As all-natural products when dispersed in water, bacteria and fungi will start to develop and induce odor and a modification of the aspect of the dispersion. These evolutions are not desirable for customers and could impact performances. It was determined that, without any thermal treatment, odor starts to develop after 3 days. Consequently, any stop of the production system for more than 2 days requires a complete system flush and cleaning.

[0309] Evaluations presented in this Example were conducted with a 20 wt % solid rapeseed micronized meal dispersion in water (prepared according to Example 2: “Grinding and thermal

treatment” at pilot scale) without stirring (at rest).

[0310] In order to quantify microbiological stability, the three tests described in Example 1 (“Raw material microbiological stability”) were implemented to follow the evolution of the odor, surface appearance and biological activity of the dispersion.

[0311] The results regarding the odor and aspect of the dispersions are presented in Table 7 and FIG. 11 respectively.

TABLE-US-00007 TABLE 7 Evaluation of the odor of dispersions of rapeseed meal J0 J1 J2 J3 J4 J5 J6 J7 J8 J9 J10 Rapeseed meal 1 4 5 5 5 without treatment Rapeseed meal 1 3 3 3 4 5 with temperature ramp up to 115° C. in pilot dryer

[0312] As observed in Table 7, the thermal treatment enables to delay odor appearance from 3 to 7 days. Moreover, surface aspect from pictures presented on FIG. 11 is better with a dispersion using thermally treated rapeseed meal. No development of fungi or mould is observed after 7 days whereas it is noticeable after only 3 days with rapeseed meal without treatment. These two observations evidence a significant improvement of odor and aspect versus time in addition to an ease of use, for the present invention.

[0313] In order to confirm these results, an industrial scale batch (300 L) has been done with thermally treated rapeseed meal at 20 wt % solid in water. To control the microbiological development, the ATP-metry method has been used and the results are shown on FIG. 12.

[0314] Development of the strong odor of dispersion using thermally treated meal starts sooner at industrial scale (5 days) than at laboratory scale (7 days). There is a strong correlation between microorganism concentration curve and odor appearance.

EXAMPLE 8: PILOT SCALE: THERMAL TREATMENT OF PELLETS AND THEN GRINDING

[0315] This Example was conducted according to the procedure described in Example 2 (pilot scale).

[0316] Temperature of 115° C. was applied on a sample of 450 kg.

[0317] Thermal treatment at pilot scale was handled by loading materiel in the dryer. Then, a temperature ramp was applied to achieve the targeted temperature. When the targeted temperature was achieved, heating is stopped to let the material cool down below 80° C. to be able to condition it safely in bags. Then the material could be ground directly (still warm) or cooled to room temperature overnight. All treatments were done under atmospheric pressure and are summarized in Table 8.

TABLE-US-00008 TABLE 8 Treatment conditions of rapeseed meal pellets Viscosity of a 25 wt % Thermal treatment Flow ground Raw Treated Temperature Grinding Particle rate rapeseed meal material Temperature mass profil conditions size kg/h dispersion Rapeseed / / / cold pellets 30 µm 80 1177 mPa .Math. s meal pellets 115° C. 450 kg heating 101 min hot pellets 30 µm 56 237 mPa .Math. s GC 12/18-06 cooling 105 min cold pellets 30 µm 92.4

[0318] When no thermal treatment is applied, mill throughput is about 80 kg/h. During the first heat test, the material was ground directly after the thermal treatment (115° C.) while pellets were still warm (about 80° C.). One can observe that the throughput is dramatically reduced to 56 kg/h. Once pellets are cooled down to room temperature (overnight) the throughput increases up to 92 kg/h.

[0319] The viscosity of dispersions prepared at 25 wt % solid content from rapeseed meal pellets treated as described above (and after cooling) was measured and is presented on Table 8. From this Table 8, decrease of viscosity can be observed when a thermal treatment is applied. It clearly demonstrates the efficiency of applying a thermal treatment on the viscosity reduction. The feasibility tests at the laboratory scale and at the pilot scale show that this treatment is able to decrease the viscosity.

[0320] Viscosity of the dispersion of ground rapeseed meal pellets has been measured at several solid contents versus the untreated material and the results are shown on FIG. 13. Once again,

thermal treatment significantly decreases viscosity of ground rapeseed meal in dispersion with water for all solid contents from 23 to 30 wt %. An acceptable viscosity from the process point of view could be get to solid content up to 27 wt % versus untreated material where the maximum acceptable solid content is 23 wt %.

[0321] The viscosity of product emulsion at two pMDI/meal weight ratios (extreme): 2 and 4 was also checked with several solid content dispersions from 20 to 30 wt %, the results being shown on FIG. 14.

[0322] One can observe that viscosity obtained with an emulsion ratio of 4 are higher than ratio 2 for each solid content and the curves look parallel in a log/lin scale. As previously observed, higher is the dry content, higher is the viscosity. The viscosity obtained here is in the same order of magnitude than laboratory treated meal even if dispersion viscosity obtained at pilot scale are higher than laboratory scale.

[0323] At the same time, emulsion quality was checked with the microscope and the results are shown on FIG. 15.

[0324] As expected, when pMDI/meal ratio is increased, droplet size of pMDI decreases at the same dispersion solid content. Another observation is the reduction of droplet size of pMDI when solid content in dispersion is increased. These emulsion aspect evolutions are consistent with viscosity variation as previously observed.

[0325] Adhesive formulations of rapeseed meal (not treated and thermally treated at 115° C.) were made, and they were added to wood particles to make wood boards. 2 Formulations were used:

[0326] First formulation (columns with diagonal lines on FIGS. 16 and 17): 1.82 part of pMDI, 0.91 dry part of ground rapeseed meal at 20 and 27 wt % solid content in water (in order to obtain pMDI/meal weight ratio=2), and 0.35 dry part of wax in emulsion for 100 parts of dry wood, the parts being weight parts. [0327] Second formulation (columns with filling points on FIGS. 16 and 17): 1.82 part of pMDI, 0.46 dry part of ground rapeseed meal at 20 and 27 wt % solid content in water (in order to obtain pMDI/meal ratio=4), and 0.35 dry part of wax in emulsion for 100 parts of dry wood, the parts being weight parts.

[0328] Water was added to the above formulations to achieve a targeted moisture content of the mat of 10% atmo. Once boards were made and conditioned, samples were cut in order to test mechanical performances (internal bonding, see FIG. 16) and water resistance (thickness swelling after 24 h in water, see FIG. 17) as described in Example 1. Thick black lines on FIGS. 16 and 17 represent average density of samples from each panel. One can observe that boards prepared from the first formulation (bars having diagonal lines, ratio 2) exhibit a good level of performance at about 0.5 N/mm². Treated and untreated are not significantly different. Boards prepared from a composition comprising untreated meal and at a ratio of 4 (bar with filling points) exhibits lower performances than boards comprising a composition wherein a thermal is applied to the meal.

[0329] Lastly, TS results at ratio 2 (bars having diagonal lines) have a good performance level versus boards made with untreated ground rapeseed meal, especially treated meal with 20 wt % solid content dispersion.

[0330] In short, mechanical performance and water resistance exhibit similar performance levels with heated treated meal versus untreated meal. It shows that thermal treatment does not negatively affect end-use performance while improving processability.

EXAMPLE 9: PILOT SCALE: GRINDING AND THEN THERMAL TREATMENT OF PELLETS

[0331] This Example was conducted according to the procedure described in Example 2 (pilot scale).

[0332] In this approach, pellets were first ground at an industrial scale, and then thermally treated into the same pilot dryer used previously. Tested parameters are the same as in Example 8. The main goal is to determine efficiency of this second approach and assess the impact of thermal treatment order. From the process point of view, this method was easier to handle (no cooling

required) and gave the highest throughput.

[0333] Viscosity curves of a dispersion of ground rapeseed meal ($d(50)=30\text{ }\mu\text{m}$) without treatment and with a thermal treatment at 115°C . were made, the solid content varying between 20 wt % and 30 wt %. The results are shown on FIG. 18.

[0334] Again, thermal treatment significantly decreases viscosity of ground rapeseed meal in dispersion with water at a comparable level than previously observed at laboratory scale from this experiment. However, it can be noted that this order is more effective than thermal treatment before grinding (see Example 8, FIG. 13).

[0335] Then, emulsions were prepared by mixing pMDI at the weight ratio pMDI/dry ground rapeseed meal of 2 and 4, with dispersions in water of ground rapeseed meal ($d(50)=30\text{ }\mu\text{m}$) with and without thermal treatment, at solid contents of 20 wt %, 23 wt %, 25 wt %, 27 wt % and 30 wt %, following the method described in Example 2 to prepare emulsions. The viscosity of these emulsions was measured and the results are shown on

[0336] FIG. 19.

[0337] As previously observed, the positive impact of thermal treatment on emulsion viscosity is confirmed. Trends are kept regarding influence of pMDI/meal ratio and solid content of the emulsions. Unlike dispersion viscosity where an impact of thermal treatment process (before or after grinding) could be observed, emulsion viscosities look similar whatever the product is treated before or after grinding.

[0338] Emulsions' quality was characterized by placing a sample of each emulsion under microscope (magnification $\times 5$) and pictures were taken with the camera. The results are shown on FIG. 20.

[0339] Trends are kept in general. For ratio 2, good enough emulsion was obtained whatever is the thermal treatment. In general, droplet size is higher when changing from an untreated meal to a treated meal. For ratio 4, it was hard to have good emulsion at solid content above 23 wt %. It could be link to the lack of volume to emulsify pMDI.

[0340] Adhesives based on these samples of rapeseed meal (not treated and thermally treated at 115°C .) were made, and they were added to wood particles to make wood boards. Two formulations were used: [0341] First formulation (columns with diagonal lines on FIGS. 21 and 22): 1.82 part of pMDI, 0.91 dry part of ground rapeseed meal at 20 and 27 wt % in water (in order to obtain pMDI/meal weight ratio=2), and 0.35 dry part of wax in emulsion for 100 parts of dry wood, the parts being weight parts. [0342] Second formulation (columns with filling points on FIGS. 21 and 22): 1.82 part of pMDI, 0.46 dry part of ground rapeseed meal at 20 and 27 wt % in water (in order to obtain pMDI/meal weight ratio=4), and 0.35 dry part of wax in emulsion for 100 parts of dry wood, the parts being weight parts.

[0343] Water was added to the above formulations to achieve a targeted moisture content of the mat of 10% atro. Once boards were made and conditioned, samples were cut in order to test mechanical performances (internal bonding, see FIG. 21) and water resistance (thickness swelling after 24 h in water, see FIG. 22) as described in Example 1. Thick black lines on FIGS. 21 and 22 represent average density of samples from each panel.

[0344] At ratio 2, treated meal exhibits better performance (IB and TS) than untreated meal. At ratio 4, all formulations exhibit a good performance level. Internal bonding of boards prepared from formulations having 27 wt % solid content and a pMDI/meal weight of ratio 4 was lower than internal bonding of boards prepared from formulations having 20 wt % solid content and a pMDI/meal weight of ratio 4 with heat treated meal.

[0345] According to these Figures, and for each formulation, the thermal treatment of ground rapeseed meal has no significant influence on mechanical performances (IB) and water resistance (TS) for boards made with the adhesives according to the invention. It confirms that thermal treatment has no negative impact on performances.

EXAMPLE 10: UPSCALE WOOD PANELS: OSB

[0346] In order to confirm these performances at a larger scale, wood tri-layers panels of 60×60 cm with strands (large particles for OSB) and wood particles were made at a pilot scale (see Example 1: Preparation of OSB). This allows testing at a larger scale with another type of wood (strands versus particles) and at a shorter press time with process condition closer to industrial standard. [0347] To form OSB, binder containing pMDI and ground rapeseed meal in dispersion at 20 wt % solid content was injected through an atomizer (spinning disc) in a Lab Coil blender in order to mix this adhesive with wood strands from WKI (40% Core/60% Face), following 2 formulas, the target moisture being determined with a moisture analyzer in “atro” mode: [0348] Core: 3.9 parts of pMDI+0.35 dry part of wax in emulsion (for 100 parts of dried wood small strands)+water for a target humidity of 7.5% of the mix, the parts being weight parts. [0349] Face: 2.48 parts of pMDI+0.71 part of ground meal dispersed at 20% in water+0.35 dry part of wax in emulsion (for 100 parts of dried wood big strands)+water for a target humidity of 12.5% of the mix, the parts being weight parts. [0350] Core: 3.12 parts of pMDI+0.89 part of ground meal dispersed at 20% in water+0.35 dry part of wax in emulsion (for 100 parts of dried wood small strands)+water for a target humidity of 7.5% of the mix, the parts being weight parts. [0351] Face: 2.48 parts of pMDI+0.71 part of ground meal dispersed at 20% in water+0.35 dry part of wax in emulsion (for 100 parts of dried wood big strands)+water for a target humidity of 12.5% of the mix, the parts being weight parts. [0352] Binders were prepared with ground rapeseed meal that were previously heat treated (at 115° C. as described in Example 8) or not. The untreated and heat treated ground rapeseed meals (“rapeseed TT”) have a granulometry d(50) of 95 µm and 80 µm respectively. Thus, 4 different binders were used.

[0353] Then, tri-layers wood boards were formed and pressed at 210° C. at a speed of 8 s/mm in order to target a density of 580 kg/m³ and a thickness of 15 mm. 2 panels per binder were made.

[0354] Once boards were made and conditioned, samples were cut in order to test mechanical performances (internal bonding, see FIG. 23) and water resistance (thickness swelling after 24 h in water, see FIG. 24).

[0355] All formulas exhibit similar level of IB. Values at pilot scale are consistent with laboratory scale data and trends.

[0356] Thermal treatment has no influence on the mechanical performances on these OSB panels. Considering the average values and the standard deviation, no difference between boards could be made.

[0357] This pilot thermal treatment demonstrates that it enables to improve processability by a dramatic reduction of viscosity. This confirms the trends already observed at laboratory scale (oven treated). Process parameters are influenced by order of treatment (heat treated before or after grinding). From a process point of view, a thermal treatment applied after grinding seems more favorable. All samples were tested in the end use application and exhibit a low or no negative impact on performances.

EXAMPLE 11: VISCOSITY OF GROUND DRY PLANT MATERIAL IN AQUEOUS DISPERSION DEPENDING ON PH OF THE DISPERSION

Preparation of Dispersions

[0358] 80 g dispersions were prepared by adding from 20 wt % to 30 wt % of untreated or thermally treated (150° C. for 30 minutes according to Example 2 at laboratory scale) ground plant material (based on the targeted solid content accounting of the residual water adsorbed in the material and determined previously with the dry content test described in Example 1) in tap water at ambient temperature in a polypropylene beaker of 100 mL. Each dispersion is mixed using a deflocculating blade (10 blades of 49 mm diameter, rod of 7 mm diameter) for two minutes, and until an homogeneous dispersion is obtained. The particle size of the used ground materials is about d(50)=30 µm. Then, the pH of the dispersion is adjusted by addition of a solution of NaOH at 5 mol/L to increase pH, or by addition of a solution of HCl at 5 mol/L to decrease pH.

Results

[0359] Viscosity of dispersions made with sunflower meal (20 wt % dispersion in water) and rapeseed meal (30 wt % dispersion in water) was measured using the rheometer (within 15 minutes maximum) as described in Example 1, and is presented on FIG. 25 for sunflower meal and on FIG. 26 for rapeseed meal.

[0360] Viscosity of dispersions made with thermally treated ground sunflower or rapeseed meal is lower than viscosity of dispersions made with the corresponding untreated meals. Thus, the thermal treatment process contributes to improve processability of these ground plant-based dispersions by decreasing viscosity, whatever the pH of these dispersions.

EXAMPLE 12: PROPERTIES OF DISPERSIONS AND ADHESIVE COMPOSITIONS OBTAINED FROM HEAT TREATED DRY PLANT MEAL ACCORDING TO THE INVENTION AND FROM PLANT MEAL HEAT TREATED IN AQUEOUS CONDITIONS

Thermal Treatment of Plant-Based Raw Materials

[0361] Soy flour (7B soy flour grade supplied by ADM, obtained by grinding soy meal) and ground sunflower meal (sunflower meal pellets ground at a granulometry of $d_{sub.50}=30\text{ }\mu\text{m}$) were heat-denatured in water and then mixed with urea (commercial grade supplied by YARA France) to produce a soy/urea aqueous product (Formula 1A in Table 9 below) and a sunflower/urea aqueous product (Formula 1B in Table 10 below), following the procedure described in paragraph of WO 2008/011455 A1: water was charged into a closed laboratory reactor IKA LR 1000 control equipped with a heating device, temperature controller and mechanical stirrer. The plant-based raw material was added to the water at room temperature over a period of three minutes. The mixture was stirred for five minutes to homogeneity and then heated to 90° C. over thirty minutes. The reaction was held at 90° C. $\pm$ 2° C. for one hour with stirring at 80 rpm. Then, the urea was added to the plant-based raw material and the mixture was held at 90° C. $\pm$ 2° C. with stirring for one hour. Then, heating was stopped to let cool the mixture and store it for use in plastic bottles at room temperature. A measurement of the solid content of these products was performed.

TABLE-US-00009 TABLE 9 Formula 1A Sequence Ingredient Total mass (g) Solid content (g) 01 Water 660.3 0 02 Soy Flour 150 143 03 Urea 143 143 Total 953.3 286 % Solid content 30

TABLE-US-00010 TABLE 10 Formula 1B Sequence Ingredient Total mass (g) Solid content (g) 01 Water 660.3 0 02 Sunflower meal 150 143 03 Urea 143 143 Total 953.3 286 % Solid content 30

[0362] In order to compare this thermal treatment in aqueous dispersion (wet thermal treatment) and thermal treatment according to the present invention (dry thermal treatment), aqueous dispersions containing thermally treated soy flour (150° C./30 min in dry condition) (Formula 2A) and thermally treated sunflower meal (150° C./30 min in dry condition) (Formula 2B) according to the present invention (dry thermal treatment at laboratory scale, Example 2, second paragraph of "Preparation of heat treated ground dry material at laboratory scale") mixed with urea were prepared to reach the same plant meal/urea ratio and the same solid content of these dispersions as respectively Formulas 1A and 1B. For all products (Formulas 1A, 1B, 2A and 2B), dry content and viscosity were measured.

[0363] In a second step, this protocol was repeated but without addition of urea. In this way, 4 aqueous dispersions with a targeted dry content of 15 wt % was prepared containing a thermally treated soy flour in dispersion (90° C./1 h in aqueous dispersion) (Formula 3A) and a thermally treated sunflower meal in dispersion (90° C./1 h in aqueous dispersion) (Formula 3B), a thermally treated soy flour (150° C./30 min in dry condition) in dispersion (Formula 4A) and a thermally treated sunflower meal (150° C./30 min in dry condition) in dispersion (Formula 4B) according to the present invention. A measurement of the dry content and viscosity of these products was realized.

Preparation of Adhesive Compositions

[0364] The pH of Formulas 1A and 2A were adjusted to reach the value of 7 with stirring by using a H.sub.2SO.sub.4 solution (1 mol/L) or a NaOH solution (1 mol/L) depending of the starting pH.

These two adhesive compositions are named respectively Formula 5A and Formula 5B. Viscosity of these two Formulas was measured.

[0365] 5% (dry weight/dry weight) of pMDI was added to Formula 5A (based on Soy flour+urea contents), and 5% (dry weight/dry weight) of pMDI was added to Formula 5B (based on soy flour+urea contents), at room temperature with agitation in a 100 ml beaker using a mixing blade and an overhead mechanical stirrer. These two adhesive compositions are named respectively Formula 7A and Formula 7B. Viscosity of these two Formulas was measured.

Protocol of Adhesion Performances Characterization

[0366] Adhesion performances of Formulas 5A, 5B, 7A and 7B were characterized following this protocol: The Automated Bonding Evaluation System (ABES) from AES, Inc. was used to measure the shear strength of the adhesive bond of the different samples as developed over time under specific pressing times/temperatures. Two ply were cut using the ABES stamping apparatus from Eastern White Pine veneer such that the final dimensions were 11.7 cm along the grain, 2.0 cm perpendicular to the grain and 0.08 cm thick. The adhesive to be tested was applied to one end of the sample such that the entire overlap area is covered, being in the range of 4-5 mg/cm² on a wet basis. The sample was then bonded to a second veneer (open time of less than 15 seconds to ensure excellent transfer) and placed in the ABES unit such that the overlap area of the bonded samples was 0.5 cm by 2.0 cm. All samples were pressed for 9 different times (10, 30, 60, 90, 150, 210, 270, 360 and 500 seconds) at 120° C., followed by a cooling step of 4 seconds thanks to pressurized air, and finally tested within the ABES unit itself within seconds after cooling.

Results and Discussions

[0367] The results of the product characterization from Formulas 1A, 1B, 2A and 2B are shown in Table 11:

TABLE-US-00011 TABLE 11 Dry content and viscosity of Formulas 1A, 1B, 2A and 2B

Formula	Dry content (%)	Viscosity (mPa .Math. s)
1A	28.3	4720
1B	29.2	11300
2A	28.3	630
2B	29.2	70

[0368] One can notice that viscosity of Formulas 2A and 2B, using dry thermally treated raw materials according to the present invention is much lower than viscosity of products made with a thermal treatment in aqueous dispersion following the protocol of the WO 2008/011455 A1 application. Thus, the present invention enables a decreased viscosity with an equivalent dry content.

[0369] The preparation procedure was repeated without addition of urea, (Formulas 3A, 3B, 4A and 4B) and results are presented in Table 12:

TABLE-US-00012 TABLE 12 Dry content and viscosity of Formulas 3A, 3B, 4A and 4B

Formula	Dry content (%)	Viscosity (mPa .Math. s)
3A	14.83	2257
3B	14.47	3685
4A	14.83	55
4B	14.47	50

[0370] The same trend was observed. The dry thermal treatment according to the present invention enables a much lower viscosity with a similar dry content of the aqueous dispersion.

[0371] Then, Formulas 5A, 5B, 7A and 7B were prepared and characterized. Values of viscosity for all Formulas are shown in Table 13:

TABLE-US-00013 TABLE 13 Viscosity of Formulas 5A, 5B, 7A and 7B

Formula	Viscosity (mPa .Math. s)
5A	3694
5B	2215
7A	4857
7B	2313

[0372] Once again, adhesive compositions formulated with thermally treated soy flour (in dry condition) according to the present invention showed a lower viscosity than thermally treated soy flour (in wet condition) according to WO 2008/011455 A1 application, (40% lower viscosity for Formula 5B compared to Formula 5A, i.e. without pMDI addition), and with a better improvement in the case of pMDI addition (53% lower viscosity for Formula 7B compared to Formula 7A, i.e. with 5 wt % pMDI addition).

[0373] Finally, the adhesion performance of these different products was characterized following the described protocol using the Automated Bonding Evaluation System (ABES). The results are plotted in FIG. 27 relative to press time to determine the relative strength development of these different adhesives as a function of time.

[0374] Formula 5B demonstrates the better adhesion properties of an adhesive obtained using a dry thermally treated soy flour according to the present invention, without pMDI addition, in comparison with an adhesive obtained using a thermally treated soy flour according to WO 2008/011455 A1 application (Formula 5A). With pMDI addition, Formulas 7A and 7B showed similar adhesion performances over time for both thermal treatments. But the thermal treatment according to the present invention enabled a much lower viscosity, as seen previously.

EXAMPLE 13: PROPERTIES OF ADHESIVE COMPOSITIONS OBTAINED FROM HEAT TREATED DRY PLANT MEAL ACCORDING TO THE INVENTION AND FROM ACID DENATURED PLANT MEAL

Acid Denatured Adhesive Preparation

[0375] According to acid treatment of US 2011/0048280 A1 patent application (Example 7 of US 2011/0048280 A1), soy flour (ground soy meal) was acid-denatured and then combined with urea to produce a stable soy/urea aqueous product (Formula 1). The formula used for this experiment is provided in Table 14:

TABLE-US-00014 TABLE 14 Formula 1 Sequence Ingredient Wet mass (g) Solids (g) 01 Water 304.3 0 02 Sodium Bisulfite 1.56 1.56 03 Soy Flour 164.5 156.3 04 Urea 156.3 156.3

[0376] Water was loaded in a 1 L beaker equipped with an overhead mechanical stirrer. The sodium bisulfite was added to the room temperature water followed by addition of the soy flour over a period of five minutes. The mixture was stirred for 30 minutes at room temperature. Sulfuric acid was then added dropwise to the rapidly stirring mixture until a pH of 3.5 was reached and, subsequently, held for an additional 30 minutes. Then, sodium hydroxide was added slowly to the rapidly stirring mixture to raise the pH to 7, and the mixture was stirred for 5 minutes. Urea was then quickly added to the rapidly stirring acid denatured soy mixture and allowed to stir for an additional 5 minutes. The adhesive was finally stored for use in plastic bottles while being maintained at room temperature. The adhesive (Formula 1) was a very homogeneous light tan, creamy product. The dry content of the final adhesive was measured as 43.77 wt %. A measurement of the viscosity was carried out.

[0377] In order to compare this acid-denatured treatment and dry thermal treatment according to the present invention, an aqueous dispersion containing thermally treated soy flour (150° C./30 min in dry condition) (Formula 2) with urea was prepared to reach the same soy/urea ratio (1:1) and the same dry content (43.77 wt %) as the adhesive prepared above (formula of Table 14). A measurement of the viscosity was carried out.

Adhesives Combined with pMDI

[0378] 67% (w/w) of pMDI was added to Formula 1 on a dry basis, and 67% (w/w) of pMDI was added to Formula 2 on a dry basis, at room temperature with agitation in a 100 ml beaker using a mixing blade and an overhead mechanical stirrer. These two adhesives are named respectively Formula 5 and Formula 6, and have a pH of about 7. Viscosity of these two Formulas was measured.

Protocol of Adhesion Performance Characterization

[0379] Adhesion performances of Formulas 1, 2, 5 and 6 were characterized following the protocol described previously (Example 12 above) using The Automated Bonding Evaluation System (ABES). All samples were pressed for 4 different times (10, 30, 60 and 90 seconds) at 120° C. and tested within the ABES unit itself within seconds after pressing.

Results and Discussions

[0380] Viscosity of Formulas 1, 2, 5 and 6 is shown in Table 15:

TABLE-US-00015 TABLE 15 Viscosity of Formulas 1, 2, 5 and 6 Formula Viscosity (mPa .Math. s) 1 5294 2 355000* 5 32840 6 1225000 *This high viscosity value results from the high dry content of the formula (43.77 wt %) and the presence of urea (dry content: 21.88 wt %) in the formula

[0381] The acid-denatured soy flour/urea adhesive of Formula 1 exhibits a lower viscosity than the

dry thermally treated soy flour/urea adhesive of Formula 2. This behavior can be explained by plant-based material modification due to acidification. Indeed, a very acidic environment enables separation of amino acids groups from protein chains. Protein chains become shorter and physical interactions become weaker. This can therefore induce an important decrease of viscosity. The same trend was observed after addition of pMDI. Viscosity of Formula 5 (with acid-denatured soy flour) is lower than viscosity of Formula 6 (with dry thermally treated soy flour).

[0382] The adhesion performance of these different products was characterized following the described protocol using the Automated Bonding Evaluation System (ABES). The results are plotted in FIG. 28 relative to press time to determine the relative strength development of these different adhesives as a function of time.

[0383] Formulas 1 and 2 showed similar adhesion performances, but these performances are too low to obtain a good adhesive without pMDI addition. On the contrary, Formula 6 demonstrates the better adhesion properties of an adhesive using a dry thermally treated soy flour according to the present invention with pMDI addition, in comparison with an adhesive using an acid-denatured soy flour with pMDI addition (Formula 5) according to US 2011/0048280 A1 patent application. Therefore, even if adhesives prepared with acid-denatured soy flour have a lower viscosity, adhesion performances are very low to obtain a good adhesive for wood-based panels. The acid-denatured treatment decreases adhesion performances of plant-based raw material because of separation of amino acids groups from protein chains. On the contrary, dry thermal treatment according to the present invention enables to obtain sufficient adhesive performances to use dry thermally treated plant-based material associated with pMDI as a wood-based panel adhesive.

EXAMPLE 14: VISCOSITY OF DISPERSIONS PREPARED FROM HEAT TREATED GROUND PLANT MEAL AND GROUND PLANT MEAL PRESSED HOT

Materials and Methods

[0384] An expeller rapeseed meal from Sanders Bretagne (St-Gérard, France) has undergone the following procedure during the crushing process: the seeds were pre-heated to approximately 115° C., ground/rolled and then pressed in a screw-press where the temperature has reached up to approximately 120° C. Finally, the rapeseed press meal is placed in a cooler to be cooled down to ambient temperature to obtain a rapeseed press flake.

[0385] In order to prove the improvement of the dry thermal treatment according to the present invention on this rapeseed press flake, this one has been ground to obtain a ground rapeseed press flake with a targeted granulometry d.sub.50 around 30 µm, by following the grinding methods described in Example 2, first paragraph of “Preparation of heat treated ground dry material at laboratory scale” (Sample A). A dry content measurement was realized, and the measured value was 96.2%.

[0386] A thermal treatment for 30 minutes at 150° C. in a lab oven was applied on a sample of the rapeseed press flake according to the present invention, by following the grinding and thermal treatment methods described in Example 2, “Preparation of heat treated ground dry material at laboratory scale” (Sample B).

[0387] To check the efficiency of the dry thermal treatment on viscosity, dispersions were prepared by adding from 20% to 40% of the native sample (Sample A) to water. The experiment was reproduced by adding from 20% to 40% of the dry thermally treated sample (Sample B) to water. Prepared dispersions were based on the targeted solid content (accounting of the residual water adsorbed in the material and determined previously with the dry content test) in tap water at ambient temperature in a polypropylene beaker of 100 mL. The dispersions were mixed using a deflocculating blade (10 blades of 49 mm diameter, rod of 7 mm diameter) for two minutes. Viscosity of all these dispersions was measured.

Results and Discussions

[0388] All viscosity values (in mPa.Math.s) of the two samples in aqueous dispersions are summarized in Table 16:

TABLE-US-00016 TABLE 16 Viscosity of aqueous dispersions prepared from samples A and B
Concentration of raw materials in dispersion with water (%) 20% 25% 30% 35% 40% SAMPLE A
No treatment 112 380 3100 46000 SAMPLE B 150° C./30 min 25 63 350 2170 40500

[0389] Dispersions prepared from sample B, consisting of water and a dry thermally treated ground rapeseed press flake according to the present invention (dry thermal treatment at laboratory scale), exhibit a dramatic reduction of viscosity in comparison with dispersions prepared from sample A (ground native rapeseed press flake at a similar solid content in the aqueous dispersion). These results show an impact of dry thermal treatment according to the present invention on rapeseed flake which already has undergone a treatment at 120° C. during the crushing process.

EXAMPLE 15: HEAT TREATED SUNFLOWER MEAL VIA A CONTINUOUS PROCESS

Equipment & Products

Products

[0390] Sunflower meal was grinded using Attrition mill (from Attritor Ltd) or Impact Mill (from Hosokawa). Independently of the grinding equipment, the granulometry was the same with a d.sub.50 of 30 µm.

Thermal Treatment EQUIPMENT

[0391] A sterilization reactor was used (FIG. 29). The ground sunflower meal was placed in a buffer hopper 1 which feeds the reactor and the dosage was performed by controlling speed of a feeding screw. The heat treatment was carried out in the horizontal reactor 2 where the product was conveyed thanks an endless screw. The reactor could be warmed using hot air or steam injection 3, a hot air or steam exit 4 being at the end of the reactor. The interest of steam is to avoid or limit product desiccation. The product was heated while it was conveyed through the reactor at a controlled speed. The residence time was between 5 and 30 minutes. The reactor could be also heated by a thermal fluid using a double jacket envelop to ensure a perfect control of temperature for heat treatment (by using thermo-couples placed inside the jacket in contact with the product). An alternative to the usage of thermal fluid is a heating system using Joule effect (electrical heating). In this case, the screw is heated, and it controls product temperature inside the reactor. At the exit 5 the product was cool down using room temperature air.

[0392] From this general description, 3 configurations have been studied: [0393] Configuration n°1: reactor with a single screw electrically heated; [0394] Configuration n°2: reactor with a single screw electrically heated and the hot air injection unit; [0395] Configuration n°3: reactor with double screw system in combination with a heated doubled jacket using oil as a thermal fluid. In this last configuration, the maximal temperature could not exceed 190° C. (limitation linked to the oil thermal stability). The oil circulates inside the screw.

Viscosity Measurement

[0396] Viscosity was measured with a viscometer EVO expert from FungiLab equipped with a disc spindle (L3). Prior to viscosity measurement, about 100 ml of dispersion, of the desired solid content, was prepared in the adequate beaker to ensure a correct immersion of the spindle until the small mark in the spindle axis. The rotation speed was set depending on the torque scale to reach a torque percentage in between 80% and 40%. With this setting, viscosity measurement higher than 3000 mPa.Math.s cannot be measured. When a dispersion had a viscosity higher than 3000 mPa.Math.s the value ">3000" was reported in the result tables (Tables 17, 18 and 19). The value was recorded after 1 min stabilization time while running the viscometer.

Results & Discussion

[0397] The results obtained on product viscosity reduction versus the thermal treatment parameter (time and temperature) and configuration (screw electrically heated in combination or not with hot air, double jacket reactor with thermal fluid) are shown below.

Configuration n°1

[0398] For this configuration an electrically heated screw was used to apply a thermal treatment on the ground sunflower meal. The flow of material was controlled to get the reactor filled at 20% of

its total capacity. Screw temperature was set between 100 to 180° C. with a residence time in between 5 to 30 min. Below 100° C., the temperature was too low to get a significant reduction of viscosity of the dispersion with a reasonable residence time. Above 180° C. it was observed char formation which could affect product performance and could be a risk of fire in the equipment. It was considered a residence time upper than 30 min prohibitive in term of flow rate and operating cost.

[0399] For each condition powder moisture was measured with a moisture analyzer in “atro” mode, right after the thermal treatment. Dispersion of product was prepared at 20 wt % solid content with tap water after product cooled down around room temperature. All results are reported in Table 17.

TABLE-US-00017 TABLE 17 Viscosity of aqueous dispersions depending on the heat treatment conditions of the sunflower meal

Screw	Residence	Flow	temperature	time	rate	Moisture	Viscosity
Test (° C.)	(min)	(kg/h)	(% atro)	(mPa .Math. s)			
Not treated	6.47	>3000	1-T1	140	20	n/a	0.74 620
1-T4	120	30	12	0.65	840	1-T5	150 30 12 0.20 61
1-T6	160	30	12	0.19	56.5	1-T7	150 20 n/a 0.39
77 1-T8	170	10	40	2.35	1050	1-T10	170 15 30 1.64 650
1-T11	180	10	30	1.56	740	1-T12	180 5 52
2.89	977						

[0400] In certain industrial use, a processable dispersion needs to have a viscosity below 500 mPa.Math.s. Thus a temperature of 150° C. appears to be required to get enough viscosity reduction in a reasonable residence time (20 min at 150° C.) in the reactor. Increasing the temperature could help to reduce the treatment time. In this case, the moisture content was dramatically decreased to a value lower than 1% which indicates that the thermal treatment was highly effective. However, a dramatic reduction of residence time was not achieved by increasing the temperature to 170° C. or 180° C. It could be linked to the configuration setting wherein, for these 2 last temperatures, the moisture content remains around 3 to 1%.

[0401] In conclusion, it was demonstrated that the reactor can be used to apply a thermal treatment to ground sunflower meal with a level of efficiency (i.e. reduction of viscosity) equal or even better compared with the lab scale test in an oven. In other word, a continuous process can be used instead of a static process.

Configuration n°2

[0402] For configuration n°2, the same equipment was used. The unique variant in the process is the addition of a hot air injection unit (800° C.) in addition to the heated screw. This additional source of calories may help to accelerate product temperature increases.

[0403] All results and test conditions were summarized in Table 18.

TABLE-US-00018 TABLE 18 Viscosity of aqueous dispersions depending on the heat treatment conditions of the sunflower meal

Screw	Residence	Flow	temperature	time	rate	Moisture	Viscosity	
Test (° C.)	(min)	(kg/h)	(% atro)	(mPa .Math. s)				
2-T5	120	30	15	1.29	500	2-T6	130 30 15 0.84 253	
2-T7	140	30	15	0.61	162	2-T8	150 30 15 0.46 110	
2-T9	150	25	21.6	0.44	198	2-T10	160 25 21.6	
0.6 150	2-T11	160	20	23.5	0.27	210	2-T12	170 15 30 0.39 400
2-T13	170	18	25	0.48	230			

[0404] With this configuration, one can observe a reduction of the minimum temperature required to get an effective reduction of viscosity to 120° C. with 30 min as a residence time. In addition to this, the increase of temperature enabled to significantly reduce the residence time to 15 min at 170° C.

[0405] In conclusion, configuration n°2, as configuration n°1, is suitable to apply a thermal treatment to ground sunflower meal using a continuous process. Configuration n°2 gives a higher level of efficiency allowing to run at a lower temperature than configuration n°1 or for a lower residence time. In this set of experiments, a flow rate of 30 Kg/h was achieved versus 12K/h for configuration n°1.

Configuration n°3

[0406] In configuration n°3, the reactor was different in comparison with configuration n°1 and n°2. The heating source comes from a double jacket system with an oil heated at 190° C. to warm the screw. In addition, the reactor was equipped with a double screw system (CWR200 from

Celsius, NL). The screw was fed to ensure the usage of 70% of the total volume. The screw was too short (2 m) to fully cure the product. Thus, the product was collected at the exit and then re-introduced inside the reactor to simulate longer residence time.

[0407] In this configuration, ground sunflower meal was introduced in the heating reactor. The product was processed in the reactor during 30 min at 190° C. which corresponds to pass n°1. After it was reintroduced in the reactor which was pass n°2 and so on.

[0408] After pass n°3, moisture content of the heat treated sunflower meal and viscosity of the resulting dispersion were measured. All results are reported in Table 19.

TABLE-US-00019 TABLE 19 Viscosity of aqueous dispersions depending on the number of passes

Moisture	Viscosity	Test	Pass no	(% atro)	(mPa .Math. s)
Not treated	X	7.43	>3000	6-19	3 1.43 46
6-20	4	0.7	41.5	6-21	5 1.21 51

[0409] After the 3.sup.rd pass, a moisture content similar to those of configurations n°1 and n°2 was measured (1.5% ATRO); viscosity of a dispersion at 20 wt % solid content has decreased significantly versus the dispersion obtained from non-heat treated ground sunflower meal. Doing a 4.sup.th and a 5.sup.th pass did not provide additional benefits. This configuration n° 3 enables to get dramatic decrease of viscosity until a minimum value around 40 mPa.Math.s. Ground rapeseed meal (Saipol, France) (D.sub.50: 30 µm) was also processed with the same configuration. The trends were similar with ground rapeseed meal as previously observed with ground sunflower meal.

CONCLUSION

[0410] In this Example, the possibility to heat treat a ground oilseed meal with a continuous process and to get a significant viscosity reduction once dispersed in water was demonstrated. The type of heating does not impact the quality of these obtained benefits (electric screw, hot air, thermal fluid).

Claims

1. A process for preparing an adhesive composition comprising the steps of: providing a heat treated dry plant meal which is capable of reducing the viscosity of a dispersion in water of said heat treated dry plant meal compared to the same dispersion in water of the non heat treated dry plant meal, and mixing the heat treated dry plant meal with water and a reactive prepolymer to provide an adhesive composition, the reactive prepolymer not being soluble in water.
2. A process for preparing an adhesive composition comprising the steps of: providing a dry plant meal heat treated at a temperature of at least 100° C., and mixing the heat treated dry plant meal with water and a reactive prepolymer to provide an adhesive composition, the reactive prepolymer not being soluble in water.
3. The process according to claim 1, wherein the temperature of the heat treatment is at least 115° C.
4. The process according to claim 1, wherein the dry plant meal comprises at least 15 wt % of proteins, on the total weight of the dry plant meal.
5. The process according to claim 1, wherein the dry plant meal has a granulometry d50 of at least 20 µm.
6. The process according to claim 1, wherein the dry plant meal is submitted to a heat treatment during at least 5 minutes.
7. A process for preparing an article with the adhesive composition prepared according to claim 1, comprising a step of contacting the adhesive composition with a lignocellulosic material to provide lignocellulosic material impregnated with the adhesive composition.
8. A heat treated dry plant meal obtained by subjecting a dry plant meal to a heat treatment at a temperature and during a treatment time sufficient to reduce the viscosity of a dispersion of said heat treated dry plant meal in water compared to the same dispersion of the non heat treated dry plant meal with water.

- 9.** The heat treated dry plant meal of claim 8, wherein the plant meal is rapeseed or canola meal and is obtained by subjecting a dry rapeseed or canola meal to a heat treatment at a temperature of at least 115° C.
- 10.** The heat treated dry plant meal of claim 8, wherein the dry plant meal is subjected to a heat treatment at a temperature of at least 150° C.
- 11.** A heat treated dry plant meal dispersion comprising a heat treated dry plant meal of claim 8 and water, and having a viscosity at 20° C. of less than 1000 mPa.Math.s.
- 12.** An adhesive composition comprising a heat treated dry plant meal according to claim 8, water and a reactive prepolymer, the reactive prepolymer not being soluble in water.
- 13.** The adhesive composition according to claim 12, the adhesive composition having a viscosity at 20° C. of less than 1000 mPa.Math.s.
- 14.** The adhesive composition according to claim 12, the adhesive composition being sprayable.
- 15.** (canceled)
- 16.** (canceled)
- 17.** The process according to claim 1, wherein the temperature of the heat treatment is at least 135° C.
- 18.** The process according to claim 1, wherein the temperature of the heat treatment is at least 150° C.
- 19.** The adhesive composition according to claim 12, the adhesive composition having a viscosity at 20° C. of less than 500 mPa.Math.s.
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